Institut Polytechnique de Paris

Master IREN - Network Industries and Digital Economy

Nested Investor Syndication Networks

Data-Driven Insights into U.S. Communities and the Evidence of Nestedness in Silicon Valley

Master Thesis

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Abstract

This work draws on network science, economic sociology and ecology to analyse U.S. venture-capital syndication patterns, once understanding how venture capitalists organise and collaborate in funding new ventures requires moving beyond firm–level analysis to examine the large–scale structure of co-investment networks.

Using more than 100k investment records from Crunchbase (covering around 16k investors and 11k companies) we construct a bipartite network linking early–stage and late–stage investors through their shared funding rounds. Then, community detection and nestedness analysis are applied to identify structural patterns in these networks.

The analysis reveals three major investor communities, among which one community (largely centred in Silicon Valley) exhibits significantly high nestedness: less-connected investors (specialists) co-invest almost with subsets of the partners of highly connected investors (generalists), like in ecology's pollinator-plant networks.

Temporal analysis shows that the nested structure develops progressively: values begin increasing after 2012, reach statistical significance in 2019, and persist thereafter. This pattern suggests a progressive evolution, but the underlying mechanisms driving this transition remain unexplored in this work. By contrast, similarly sized communities without nested organisation display lower transaction volumes and lack geographic concentration.

These findings suggest that network topology, rather than community size or investor centrality alone, influences investment efficiency and access to capital. By integrating concepts from finance, social network theory and ecology, the thesis contributes a novel framework for analysing innovation ecosystems and opens avenues for predicting investment patterns and designing policies to foster venture-capital markets behavior.

Keywords: *venture capital, syndication networks, nestedness, network theory, investor communities, Silicon Valley, innovation ecosystems, bipartite networks, community detection, critical transitions*

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1 Introduction

The practice of investors jointly funding the same venture is known as syndication. It plays a central role in innovation financing by allowing investors to pool expertise, share risk and signal project quality to the market [1, 2].

Despite its practical importance, most empirical studies analyse syndication at the firm or deal level and provide limited insight into the system-level patterns through which investors coordinate [3, 4]. Understanding how these patterns emerge requires a multidisciplinary perspective encompassing finance, sociology, network science and ecology.

This collaborative behavior also reflects information asymmetries and screening challenges inherent in investment markets [1], as in incomplete information environments, investors rely on social relationships to reduce uncertainty and improve screening efficiency.

In venture capital markets (innovation financing included), syndication serves as a coordination mechanisms used by Venture Capitalists (VCs) to operate in complex networks where investment decisions are embedded within social structures, geographic clusters, and institutional relationships.

This study aims to apply Network Science theories and methods to analyze these coordination patterns in the syndication network of early and late stage investors operating in the U.S. market, revealing how investors organize into distinct network communities with different structural properties and organizational patterns that may influence capital allocation efficiency, deal flow, and market access.

Special attention is given to the fact that a community of prominent VCs in Silicon Valley demonstrates nested hierarchical organization patterns that develop progressively over time, with nestedness values beginning to increase after 2012 and reaching statistical significance around 2019, while other communities maintain more random, hard-to-define partnership structures. The specific factors underlying this progressive structural evolution remain unexplored in this work, presenting opportunities for future research into the mechanisms driving hierarchical organization in innovation networks [5, 6, 7].

The following sections examine the theoretical foundations for syndication behavior, drawing from literature in innovation financing, social embeddedness, network theory, and ecological systems to build a comprehensive framework for understanding syndication investments networks organization.

Nevertheless, implications on market efficiency, geographic clustering effects, and the role of innovation hubs in shaping investment ecosystems is discussed, as well as future reasearch opportunities are presented.

1.1 Syndication in Innovation Financing

Theoretical foundations for syndication behavior comes from literature on screening and information sharing among venture capitalists [2]. Syndication became relevant in that context because, in markets characterized by incomplete information (e.g. innovation financing), the decision to syndicate and the choice of syndication partners are aspects that influence both investment outcomes and network position [2].

In innovation financing, when investors face uncertainty about startup quality and potential, syndication serves multiple strategic functions: it enables knowledge pooling from partners with complementary expertise [1], reduces individual exposure to high-risk ventures, and provides valuable signals [8] about investment quality through the revealed preferences of co-investors.

In this context, considering that screening innovative ventures is frequently a multi-stage process (venture capitalists define formal pipeline which startups and often their founders need to pass through), syndication plays an important role to reduce the effort and possibilty the number of screening stages, with the cost of disclousring information, as investors are essencially competitors.

1.2 Economic Sociology and Embeddedness

While the screening and signaling functions of syndication explain why investors collaborate, understanding how these partnerships form and persist requires insights from economic sociology.

Economic sociologists have demonstrated that economic action is fundamentally embedded in ongoing social networks [9]. For instance, entrepreneurs and venture capitalists develop recurring partnerships built on trust and reputation rather than purely transactional relationships [10]. This embeddedness perspective reveals that syndication patterns emerge from social structures that extend beyond individual deal characteristics.

Central to this framework is Burt's concept of structural holes [11], which shows how investors who bridge otherwise disconnected groups gain access to unique information and control over resource flows (advantages that directly complement the screening benefits discussed earlier).In that sale manner, geographic clustering in innovation hubs such as Silicon Valley facilitates the repeated face-to-face interactions that strengthen these social ties, creating the relational foundation upon which systematic syndication patterns can emerge.

Furthermore, embeddedness creates informal hierarchies of influence and reputation within investor communities. Rather than operating as independent agents, investors form interconnected networks that share information, coordinate investment strategies, and establish social rankings based on past performance and network position [9]. These social hierarchies can evolve into the structural hierarchies measured through, for instance, nestedness analysis.

The embeddedness perspective helps explain why certain investor communities develop systematic organizational patterns while others remain more randomly structured. Social relationships, trust networks, and geographic proximity may facilitate the emergence of hierarchical organization by creating conditions where, for example, specialists naturally align with generalists' investment strategies (not purely random). This social foundation provides the context for understanding how network topology emerges from underlying social processes rather than purely market-driven mechanisms.

1.3 Network Science and Syndication Networks

In this work, we use Network Science as an umbrella term that brings together (i) network theory (how positions and ties affect outcomes), (ii) the theory of networks (how structures arise), and (iii) the graph-theoretic foundations and metrics that make these analyses possible.

Network theory (i) looks at how mechanisms and processes operating through a given network produce outcomes for people and groups. Using [12]'s distinction, it focuses on, for instance, what happens when someone has many connections, sits in a central position, or bridges others. The core question is: given a structure, what consequences follow for actors and collectives?

By contrast, theory of networks (ii) asks about the causes of structure—the antecedents of network properties in Brass's terms. It models who connects to whom, what kinds of ties they form, who becomes central, and which whole-network patterns appear (such as high centralization or small-world path lengths). The core question here is: what processes make the network take the shape it has?

Under both views sits graph theory (iii), the math of networks. It gives precise definitions of nodes and edges, paths and distances, and the core measures we use (degree, centrality, clustering, connectivity). These tools let us describe a network's shape, test ideas about causes and effects, and compare cases. Network Science (as a broader field) builds on this math to study system-level patterns, and statistics helps with modeling and uncertainty.

In syndication networks, investors are nodes and shared funding rounds are edges, while network measures quantify centrality, detect community structure and reveal degree heterogeneity [13, 14]. Many real-world networks, including VC ecosystems, exhibit heavy-tailed degree distributions generated by preferential attachment dynamics [6], and so they display small-world characteristics, combining local clustering with short global path lengths [15].

In the following sub-sections we explore concepts spaning all of those fields to set the theorical basis to our reasoning on the syndication networks construction and analysis for further chapters.

1.3.1 Unipartite and Bipartite Networks

Understanding the distinction between unipartite and bipartite network representations is fundamental to our analysis of investor syndication patterns, as it determines our ability to identify hierarchical structures like nestedness that emerge from role differentiation between network participants.

A unipartite network has a single set of nodes, and any two nodes may be linked (e.g., people connected to people). In the other hand, a bipartite network has two disjoint sets of nodes that play different roles, and links are allowed only between the sets (e.g., plants–pollinators [16] or early–late investors), never within a set. In summary, bipartite models are useful when role separation matters. Figure 1 provides an illustration of both types of networks.

These network types differ fundamentally in their matrix representations: unipartite networks are represented by square matrices where all nodes appear in both rows and columns (known as adjacency matrix), while bipartite networks use rectangular matrices where rows represent one node group and columns represent the other (biadjacency matrix), as illustrated in Figure 1.

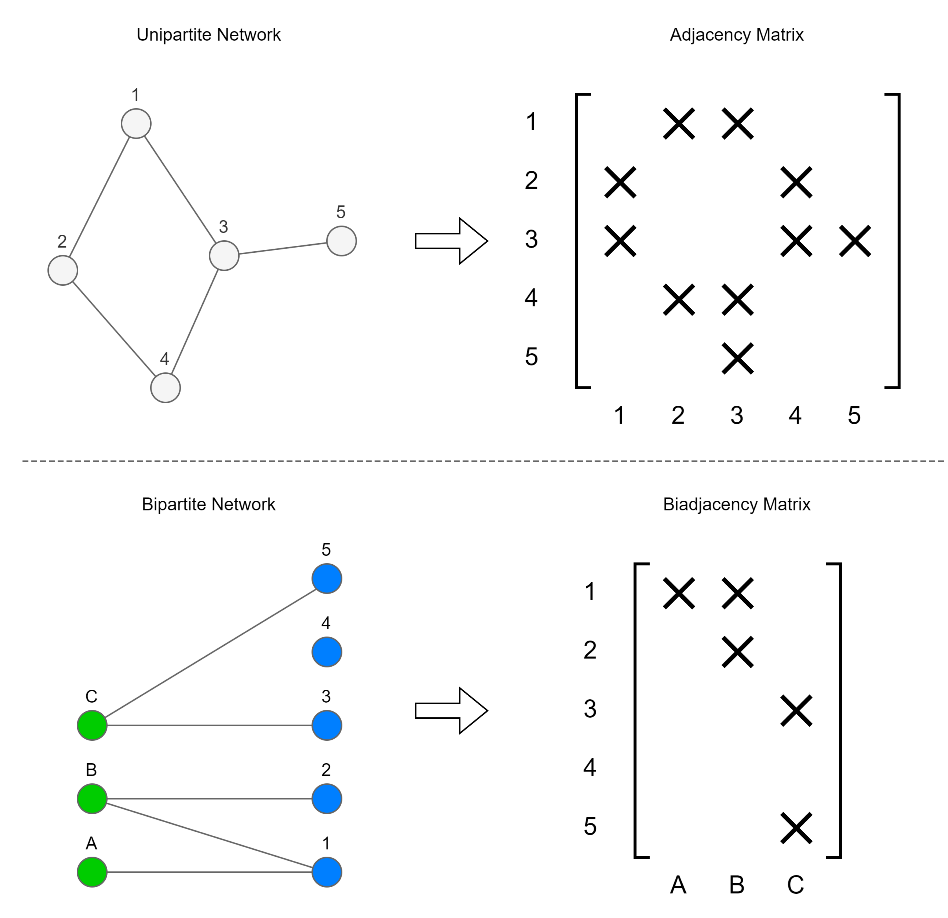


Figure 1: Top, a unipartite network—one set of nodes (white) where edges can connect any pair. Bottom, a bipartite network—two disjoint sets (green vs. blue) with edges only across sets and none within a set. Respective adjacency and biadjacency matrix on right side, where "X" is actually substituted by "1" for calculations.

This matrix structure is important because nestedness patterns (characterized by the triangular arrangement of non-zero entries when ordered by degree) can only be properly identified in the rectangular bipartite matrix format. Moreover, ecological concepts like nestedness (which we will employ to analyze investor hierarchies) are most naturally defined and measured in biadjacency matrices.

In our context, modeling investors and companies as distinct node sets in a bipartite framework allows us to capture the role-specific interactions that generate nested patterns, whereas traditional unipartite investor-investor projections [17] would obscure these hierarchical relationships.

As we will demonstrate in the methodology section and through ecological analogies, the bipartite representation preserves the structural information necessary to transpose interpretations from mutualistic ecological systems to economic and social systems.

Last but not least, in this work we explore how bipartite networks can be projected back into unipartite networks by connecting nodes from the same set whenever they share connections to the other set. This projection mechanism provides a bridge between our role-aware bipartite model and classical syndication networks, enabling compatibility with existing research while preserving distinct investor roles.

1.3.2 The Classical Syndication Model

The classical approach to studying venture capital syndication treats all investors as a single group and connects them whenever they participate in the same funding round [17]. In this model, investors are represented as nodes in a network, and an edge is drawn between any two investors who have co-invested in at least one company together.

This traditional view creates a unipartite network where, for instance, if Investor 1, Investor 2, and Investor 3 all participate in funding Company E, then it creates direct connections between 1-2, 1-3, and 2-3, as illustrated in Figure 2. The resulting network shows the web of relationships among all investors based on their shared investment activities.

While this approach has been widely used in venture capital research [17], it treats all investors identically regardless of their roles in the investment process. Early-stage investors who specialize in seed funding are connected in the same way as late-stage investors who focus on growth capital, even though their investment strategies and market positions may be fundamentally different.

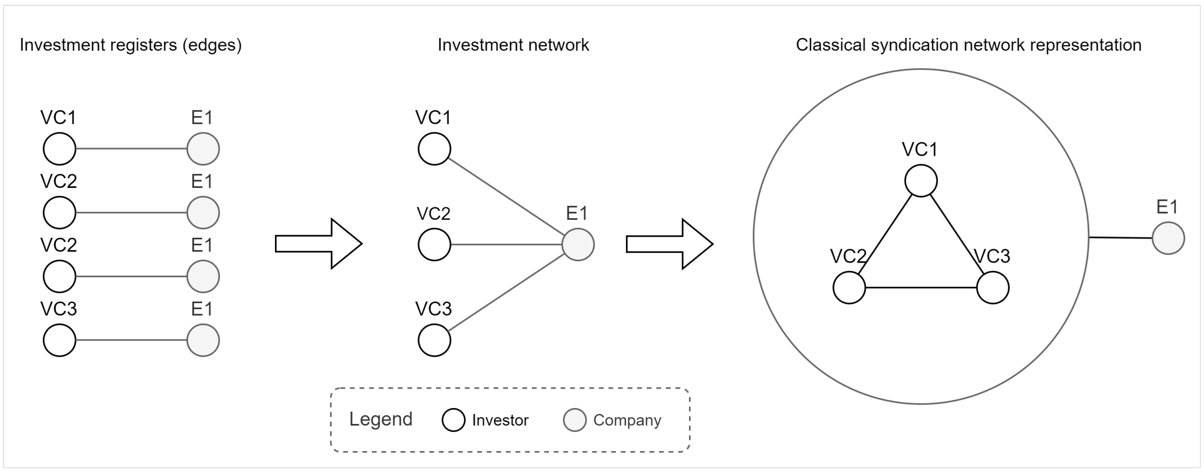


Figure 2: Classical syndication network: investors are connected whenever they co-invest in the same funding round. This unipartite view aggregates all investors into a single set, obscuring role differentiation and hierarchical patterns that may be present in the underlying bipartite structure.

1.3.3 Bipartite Early-Late Syndication Model

The classical syndication model described above treats all investors the same way, but in reality, investors play different roles depending on when they invest in a company's lifecycle. Our approach (detailed in our methodology) recognizes this by separating investors into two groups: early-stage investors (who fund seed and Series A rounds) and late-stage investors (who fund Series B and beyond).

This creates an early-late stage syndication model represented as a bipartite network where early-stage investors on one side connect to late-stage investors on the other side whenever they have both invested in the same company.

The connection between our bipartite model and the classical approach lies in network projections. A projection transforms a bipartite network back into a unipartite network by connecting nodes from the same side when they share connections to the other side, as illustrated in Figure 3.

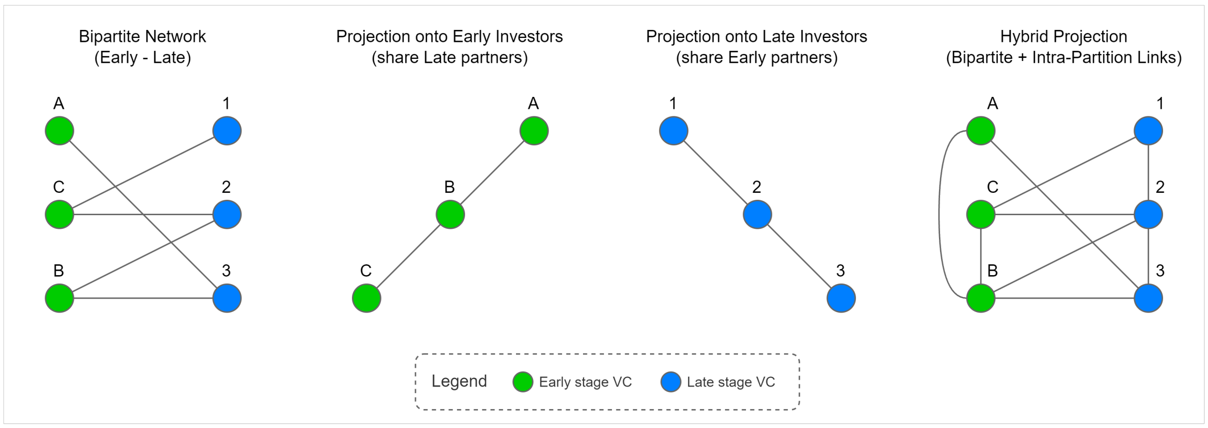


Figure 3: Illustration of bipartite network projections. The left panel shows a bipartite graph with two disjoint node sets (e.g., Early and Late investors). The middle and right panels show the corresponding one-mode projections onto each set, where edges represent shared partners in the opposite set. This process preserves role information in the bipartite base while enabling standard network analyses in the projected graphs.

Mathematically, we represent our bipartite network with a biadjacency matrix $\mathbf{B}$, where $B_{uv} = 1$ if early-stage investor u is connected to late-stage investor v. The projection onto the early-stage investors is:

$$\mathbf{W}^{Early} = \mathbf{B}\mathbf{B}^{\top}$$

where W_{ij}^{Early} counts how many late-stage partners two early-stage investors share. Similarly, the projection onto late-stage investors is $\mathbf{W}^{Late} = \mathbf{B}^{\top}\mathbf{B}$.

This projection mechanism provides a bridge between our role-aware bipartite model and classical syndication networks. When we project our Early-Late model, we recover something similar to classical investor-investor networks, even though this work does not explicitly model and explore this dynamic.

In summary, our bipartite approach enables the study of hierarchical patterns like nestedness (something obscured in classical models) while maintaining compatibility with existing venture capital research.

1.3.4 Centrality, Community Structure, and Degree Patterns

Three network concepts are particularly relevant for understanding venture capital ecosystems. First, network position and centrality metrics often quantify an investor's importance within the overall system. Highly central investors often function as gatekeepers who control information flow and access to investment opportunities [18].

In fact, different centrality measures capture distinct aspects of influence: degree centrality identifies investors with many direct partnerships, while betweenness centrality identifies those who bridge otherwise disconnected investor groups.

Second, community structure reveals how investors cluster into distinct groups with denser internal connections than external ones [19]. These communities often reflect shared investment philosophies, geographic proximity, or sector specialization.

In our case, we detect communities by maximizing *modularity*, which scores a partition by comparing the observed number of within-group edges to what a degree-preserving random network would produce. In its common form,

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

where A_{ij} is the adjacency matrix, k_i is node degree, m is the number of edges, and $\delta(c_i, c_j) = 1$ if i and j are in the same community. Louvain and Leiden algorithms search for partitions with high Q [20, 21]. This links naturally to the rest of our analysis: central and bridging investors can shape community boundaries, and modularity helps summarize heterogeneous, heavy-tailed connectivity into coherent groups for further study.

Lastly, the concept of power-law degree distributions characterize the heterogeneous heavy-tailed connectivity patterns in venture capital networks [17], which basically describes systems where a small number of investors (hubs) maintain extensive partnership networks while most participants have relatively few connections.

Power-law distributions arise from preferential attachment mechanisms, where well-connected investors attract disproportionately more new partnerships, reflecting reputation accumulation and resource concentration dynamics.

Many real-world networks, including VC ecosystems, exhibit heavy-tailed degree distributions generated by preferential attachment dynamics [6], and they display small-world characteristics, combining local clustering with short global path lengths [15].

Further, empirical evidence shows that venture capital networks consistently exhibit these power-law properties [22], where hub investors facilitate deal flow and information diffusion throughout the ecosystem, while also creating potential bottlenecks that may limit access for entrepreneurs outside established networks.

This structural feature creates conditions for the emergence of hierarchical organization patterns measured through nestedness analysis.

The methodology section builds directly on these foundations by implementing community detection algorithms that identify investor clusters, while the results section examines how community structure relates to geographic distribution, investment patterns, and hierarchical organization.

1.4 Ecology and Pollinator-Plant Networks

The power-law degree distributions discussed above provide important insights into network structure, but they cannot fully capture the hierarchical organization patterns that emerge within investor communities.

Ecology theory offers additional insight into these organisation of complex networks. In special, studies of mutualistic pollinator-plant networks have shown that interactions often form nested structures in which specialist pollinators (those that visit few plant species) typically interact with subsets of the same plants as generalist pollinators (those that visit many species) [5, 23].

1.4.1 Understanding Nestedness

To make the notion of nestedness concrete, consider a simple pollinator–plant system. Plants and pollinators form a bipartite graph where generalist species interact with many partners and specialists interact only with subsets of those partners.

When the rows and columns of the corresponding adjacency matrix are ordered by decreasing degree, the non-zero entries occupy the upper-left corner, producing a characteristic triangular pattern.

Figure 4 contrasts this bipartite representation with a standard unipartite network in which all nodes belong to the same set. The pollinator–plant analogy highlights why bipartite representations are essential for identifying hierarchical relationships like nestedness that would be obscured in one-mode projections (unipartite networks).

Quantifying nestedness in bipartite networks has been formalised through the NODF metric [24] and related statistical tests [16, 25]. Nested structures can enhance robustness by allowing specialists to persist through their links to generalists, but they also concentrate

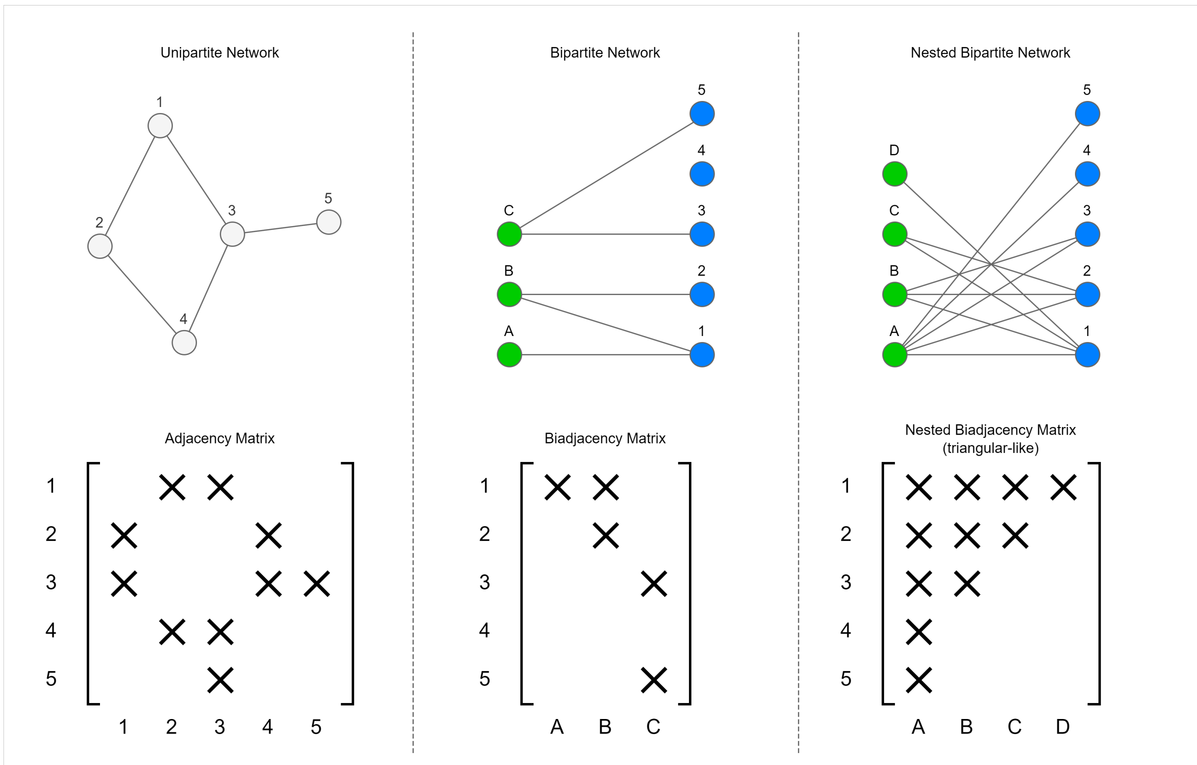


Figure 4: Illustration of nestedness in a bipartite pollinator–plant network. Panels show, from left to right, an ordinary unipartite network, a bipartite network and a nested bipartite network together with their matrix representations. Specialists connect to subsets of generalists' partners, producing the triangular pattern in the bipartite matrix.

influence in hubs and can increase vulnerability to targeted attacks [7, 26]. Recent work has explored nestedness in socio-economic systems [5], but its relevance for venture-capital networks remains largely unexplored.

1.4.2 Socio-Economic Implications of Nestedness

In socio-economic systems, several social and economic implications are associated with nestedness, as literature defends that nested structures influence competition and overall vunerability in the ecosystems. In ecology, such aspects are analog to the propensity to biodiversity (lower entry barriers) and the dilema of "stability against random species extinction" (small players) versus "vulnerability to targeted removal of highly connected species" (hubs).

Transposing to venture capital networks, nested behavior would mean that investors with fewer partnerships typically co-invest with a subset of the same companies that highly connected investors choose. This creates hierarchical organization where less connected investors operate within the partnership networks established by more connected ones.

This hierarchical arrangement has important implications for information flow and market access. In nested networks, information and opportunities tend to flow from generalist nodes (highly connected investors) to specialist nodes (less connected investors), creating asymmetric relationships that can influence deal flow, startup access to capital, and overall market efficiency [5].

Nestedness patterns are particularly relevant for understanding venture capital ecosystems because they can reveal whether certain investors function as gatekeepers who control access to broader investment networks [18]. This has direct implications for startup fundraising success and the overall efficiency of capital allocation within innovation ecosystems [19].

1.5 Research Questions and Contributions

Existing VC research has shown that syndication improves deal selection and venture performance [27, 28], yet the network-level organisation of investor communities and its implications for financing remain unclear [3, 4]. In particular, it is not known whether co-investment networks exhibit hierarchical nestedness akin to ecological systems, how such structures relate to geographic clustering, and when they arise during the evolution of the ecosystem.

To guide the analysis, the following research questions are posed:

1. Do venture-capital syndication networks in the United States exhibit hierarchical nested structures similar to those observed in ecological mutualistic networks?
2. How does geographic clustering, especially in Silicon Valley, influence the emergence and characteristics of nested structures in investor communities?
3. What are the temporal dynamics of nestedness in co-investment networks? Do hierarchical structures emerge gradually or through phase transitions as the network grows and becomes denser?

By answering these questions, the thesis seeks to illuminate how network topology shapes innovation financing and to provide a foundation for policy interventions aimed at fostering inclusive and resilient investment ecosystems.

The empirical analysis reveals three key contributions to the literature: First, we demonstrate significant heterogeneity in organizational structures across investor communities on US, with some exhibiting pronounced hierarchical patterns while others maintain more distributed partnership networks. Second, we identify a strong relationship between geographic clustering (particularly within Silicon Valley) and the emergence of nested organizational structures that confer measurable advantages in transaction efficiency and capital deployment. Third, we provide quantitative evidence that network topology, rather than community size alone, serves as a critical determinant of investment ecosystem performance.

The remainder of this paper is organized as follows. The *Methodology* section establishes our analytical framework by detailing the Crunchbase data preprocessing procedures, the construction of bipartite early-late stage investor networks, and the implementation of community detection and nestedness measurement techniques. The *Results* section first provides a comprehensive dataset characterization including syndication trends and investment activity patterns, then presents the identification and analysis of investor communities through modularity optimization, examines their geographic distributions and funding behaviors, and finally quantifies nestedness patterns using NODF metrics and statistical significance testing. The *Discussion* section interprets these empirical findings by exploring the implications of hierarchical organization for investment efficiency, analyzing the temporal dynamics of progressive nestedness development, and examining the relationship between geographic clustering and network topology. Finally, the *Conclusion* synthesizes our key contributions, acknowledges study limitations, and outlines directions for future research into network-driven innovation financing mechanisms.

2 Methodology

2.1 Data Cleaning and Preprocessing

This study uses data from Crunchbase, a commercial database that has emerged as a primary source for entrepreneurship and venture-capital research [29]. The platform contains detailed information on startups, investors and funding rounds. Its data are collected through user submissions, web crawling and partnerships, supplemented by editorial curation [29]. Crunchbase provides comprehensive coverage of U.S. technology-oriented transactions, making it well suited to studies of innovation ecosystems.

We queried Crunchbase for all investment transactions into enterprises domiciled in the United States and recorded all investors who participated in these rounds. Foreign investors are included when they invest in U.S. companies. Although Crunchbase has excellent coverage of technology-driven ventures, several limitations and potential biases must be acknowledged [29]:

- **Geographic bias**: Stronger coverage of US and Western European markets compared to emerging economies.
- **Sectoral bias**: Emphasis on technology and internet companies, potentially underrepresenting traditional industries.
- **Venture capital bias**: Better documentation of VC-backed companies compared to bootstrapped or debt-financed ventures (which may not be problematic for this study).
- **Size bias**: Larger and more successful companies are more likely to be thoroughly documented.

- **Temporal bias**: More recent information tends to be more complete and accurate than historical records.

These biases do not invalidate research using Crunchbase but require careful consideration in study design and interpretation. For network analysis of venture-capital co-investments, the VC bias may enhance data quality by focusing on the target population of interest.

The data preprocessing and cleaning follow established methodologies from entrepreneurship literature and network-analysis practice [22]. After downloading the raw data, we performed the following steps:

1. **Company data cleaning**: Removal of companies with incomplete essential information (missing unique identifiers, names, or founding years), exclusion of companies founded after 2017 to allow sufficient time for investment patterns to emerge, and removal of companies with exit status (closed, acquired, or IPO).

2. **Investment data cleaning**: Removal of investment records with missing essential linkage information (company or investor identifiers), elimination of investments with invalid funding amounts (negative or zero values), and validation of funding consistency by excluding rounds where the sum of individual investments does not match the total funding amount reported in the database.

3. **Funding threshold**: Restrict the sample to companies that raised more than $150,000 in total funding to focus on substantive investment relationships and ensure statistical reliability.

4. **Data consistency validation**: Final consistency checks to ensure all investment records reference existing companies in the dataset and all retained companies have at least one valid investment record.

2.2 Network Construction

The network construction process transforms the cleaned investment data into a structured bipartite representation. Rather than projecting a one-mode investor–investor network, we model the system as a bipartite graph connecting early-stage investors to late-stage investors through their shared portfolio companies. This approach captures both the sequential flow of capital across funding stages and the solidarity of syndication partnerships. It follows the conceptual distinction between flow and bond models in social-network analysis [13].

A one-mode projection would obscure stage-specific roles and over-estimate connectivity; the bipartite structure retains the heterogeneity of investor types and allows us to study how early and late investors are coupled through co-investment.

The construction process involves three sequential transformations, each representing a different conceptualization of network ties, each one described in 5 and in the subsequent paragraphs below.

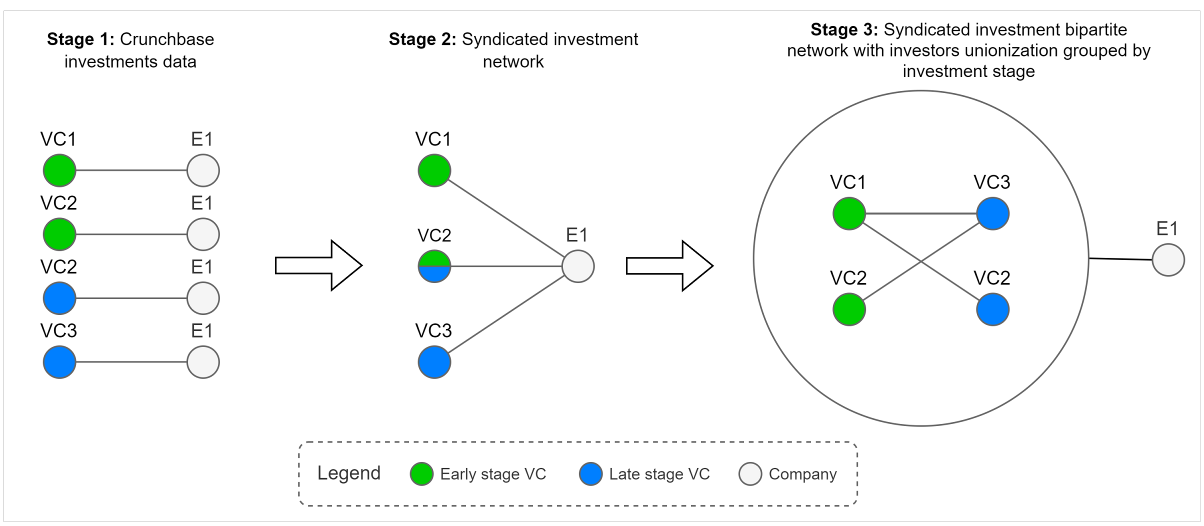


Figure 5: Pipeline from event-level records to the bipartite syndication network linking late- to early-stage investors.

Stage 1 – Raw event-type investment ties: The initial Crunchbase dataset records discrete transactional relationships between venture-capital firms and portfolio companies. These event-type ties represent what [13] characterise as nominalist network constructions: the investment transaction is the fundamental tie type at this stage. Each investment event creates a direct dyadic relationship between a venture-capital firm and a company, but these isolated ties do not yet reveal broader co-investment patterns.

Stage 2 – Syndicated investment network construction: The raw event data are aggregated to identify co-investment relationships, where multiple venture-capital firms participate in the same funding round or invest in the same company across different rounds. This aggregation transforms discrete events into a network structure in which venture-capital firms become indirectly connected through their shared ties to portfolio companies. Following [13], this step creates pathways between actors that enable analysis of their relative positions, including measures of centrality and brokerage within the investment ecosystem.

Stage 3 - Bipartite network segmentation by investment stage: The final transformation reorganizes the syndicated network into a bipartite structure that separates investors based on their participation in different funding stages.

First, investment stages are categorized into two main groups:

- Early stages: angel, pre-seed, seed, and Series A
- Late stages: Series B through Series I

This bipartite structure simultaneously reflects flow and bond models [13]. From a flow perspective, it allows us to analyse how resources and information traverse funding stages; from a bond perspective, it captures the cohesion of co-investor coalitions within each stage. Modelling both perspectives explicitly is essential because early–stage and late–stage investors perform distinct roles in syndication.

An important consideration in this construction involves investors who participate in both early and late stages (for example, a venture capital firm that invests in both Series A and Series C rounds). In such exceptional cases, the same investor appears as distinct agents on both sides of the bipartite network.

This separation is achieved by appending the investment stage identifier to the unique investor UUID during the network construction process, effectively creating stage-specific identities for multi-stage investors. This approach ensures that the bipartite structure remains mathematically valid while preserving the analytical ability to study how the same investor behaves differently across investment stages.

The resulting bipartite graph connects two distinct sets of investors: those participating in early-stage rounds and those participating in late-stage rounds. Edges represent co-investment relationships where early-stage and late-stage investors have both invested in the same company, creating a bridge between different phases of the venture capital investment cycle.

Formally, the bipartite graph $G = (U \cup V, E)$ consists of:

$$U = \{u_1, u_2, \ldots, u_m\} \text{ (late-stage VCs)} \tag{1}$$

$$V = \{v_1, v_2, \ldots, v_n\} \text{ (early-stage VCs)} \tag{2}$$

$$E \subseteq U \times V \text{ (co-investment relationships)} \tag{3}$$

To prevent spurious connections from related entities, investor pairs where the first five characters of their names match are filtered out, reducing the likelihood of including different funds from the same parent organization (even though it does not change our results as per numerous experiments). Furthermore, investors that participated in both early and late stages receive a suffix so they can be treated as distinct agents for each phase.

2.3 Community Detection

Community structure in the bipartite network is identified using modularity optimisation. Modularity was introduced by Newman as a measure of the excess density of edges within communities compared with a random null model [14]. For bipartite graphs, Barber [30] proposed an adjusted modularity function that accounts for the two disjoint node sets. We adopt a greedy optimisation algorithm that iteratively merges communities to maximise this bipartite modularity.

Greedy heuristics have proved effective at uncovering meaningful structure while scaling to large graphs [20]. Although more advanced methods such as the Leiden algorithm offer improvements in connectivity [21], we verified that applying Leiden to a subset of the data yielded similar community assignments. This robustness check suggests that our results are not an artefact of algorithm choice.

For a bipartite network, the modularity Q is defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{4}$$

where A_{ij} is the adjacency matrix, k_i is the degree of node i, m is the total number of edges, c_i is the community of node i, and $\delta(c_i, c_j)$ is 1 if nodes i and j are in the same community, 0 otherwise.

To test the stability of our findings, we repeated the community-detection procedure under multiple random seeds and compared the resulting partitions. We also performed a sensitivity analysis using the Leiden algorithm [21]. In all cases, the three largest communities remained almost identical in composition and size. This robustness gives confidence that our findings are not sensitive to algorithmic or stochastic variation.

Modularity-based community detection has proven effective in venture-capital research. Bubna et al. showed that modularity optimisation on co-investment networks uncovers persistent communities with distinct investment strategies [31]. We build on this basis to explore structural patterns that go beyond degree distributions.

2.4 Nestedness Analysis

Nestedness is a structural property observed in ecological networks [23, 24] that describes the tendency for specialists to interact with subsets of the partners of generalists. In the context of venture-capital syndication, nestedness would mean that investors with few ties tend to co-invest with a subset of the partners of highly connected investors.

We measure nestedness using the NODF (Nestedness based on Overlap and Decreasing Fill) metric [24]. NODF has become a standard measure for quantifying nested patterns in bipartite networks; it accounts for both the overlap of connections and the degree differences between nodes [16]. A perfect nested structure yields NODF = 1, while a random structure yields values close to zero.

For a bipartite adjacency matrix M with rows and columns sorted by decreasing degree, NODF is calculated as:

$$NODF = \frac{NODF_{rows} + NODF_{columns}}{2} \tag{5}$$

where:

$$NODF_{rows} = \frac{100}{R(R-1)/2} \sum_{i=1}^{R-1} \sum_{j=i+1}^{R} \frac{|N_i \cap N_j|}{k_j} \text{ if } k_i > k_j \tag{6}$$

$$NODF_{columns} = \frac{100}{C(C-1)/2} \sum_{i=1}^{C-1} \sum_{j=i+1}^{C} \frac{|N_i \cap N_j|}{k_j} \text{ if } k_i > k_j \tag{7}$$

Here, R and C are the number of rows and columns, N_i represents the set of connections for node i, and k_i is the degree of node i.

Using this method, NODF values range between 0 and 1 (perfect nestedness).

2.5 Statistical Significance Testing

To determine whether observed nestedness values differ significantly from what would be expected by chance, we employ a null-model approach using the Curveball algorithm [32]. Curveball generates randomized matrices that preserve the degree sequence of both node sets while randomising the connection patterns.

Preserving the degree sequence is critical because degree heterogeneity alone can induce apparent nestedness; a degree-preserving null model allows us to test whether the observed hierarchical organisation goes beyond what could arise from simple degree sequences [16, 25].

For each community we generate 1,000 null matrices using 10,000 Curveball iterations. Increasing the number of null matrices improves the precision of the p-value estimates at the cost of higher computational time. We verified that results were qualitatively unchanged when using 500 or 2,000 null matrices.

For each community, we compute the mean (μ_{null}) and standard deviation (σ_{null}) of the null NODF values. We then calculate a standardized Z-score to quantify how many standard deviations the observed nestedness differs from the null expectation:

$$Z = \frac{NODF_{observed} - \mu_{null}}{\sigma_{null}} \tag{8}$$

where μ_{null} and σ_{null} are the mean and standard deviation of the null distribution, respectively.

The empirical p-value is calculated as the proportion of randomised matrices that exhibit nestedness equal to or greater than the observed value:

$$p = \frac{1 + \sum_{i=1}^{N} I(NODF_{null,i} \geq NODF_{observed})}{N + 1} \tag{9}$$

where N is the number of null matrices, $NODF_{\text{null},i}$ is the nestedness score of the i-th null matrix and $I(\cdot)$ is an indicator function equal to 1 when the condition is true and 0 otherwise. Adding 1 to the numerator and denominator yields a conservative p-value estimate and avoids values of exactly zero.

The p-value represents the probability of observing nestedness as high as or higher than the observed value under the null hypothesis of random co-investment patterns. Communities with $p < 0.05$ are considered to have significantly high nestedness (when the observed value exceeds the null expectation) or significantly low nestedness (when the observed value falls below it).

Z-scores and p-values provide complementary information: the Z-score quantifies how many standard deviations the observation lies from the null mean, whereas the p-value gives the tail probability.

As an additional robustness check we repeated the null-model analysis across multiple random seeds and confirmed that the significance outcomes were stable. We also varied the threshold for community size (150 investors) and found that the qualitative conclusions about which communities exhibited significant nestedness remained unchanged.

3 Results

3.1 Dataset Characterization

The raw dataset contained information on 22,527 U.S. companies, 38,843 investors and 147,832 investment events. After the cleaning process described in the *Methodology* section, 104,618 investment records remained, involving 16,932 companies and 11,279 investors. Total capital deployed during 2004–2024 exceeded US$3.19 trillion. This cleaned dataset forms the basis for network construction and analysis.

3.1.1 Syndication Trends

Syndicated investments account for a large share of activity, as shown in Figure 6. The prevalence of syndication increased steadily during the 2000s and peaked in the early 2020s before moderating slightly. This trend reinforce the importance of studying co-investment patterns rather than isolated deals.

3.1.2 Investment Activity

Figure 7 presents the overall investment activity trends, while Figure 8 provides a detailed breakdown of temporal patterns across different investment characteristics.

From 2004 to 2024 the number of unique investors increased by more than 500% and total investment volume by over 600%. Investment counts and funding amounts do not always move in tandem: transaction volumes peaked in 2021 (18,741 deals) whereas average round sizes peaked a year earlier. These temporal variations likely reflect macroeconomic cycles, risk appetite and the maturation of venture capital as an asset class. Importantly, shifts in investment activity correspond to changes in network topology, as discussed later.

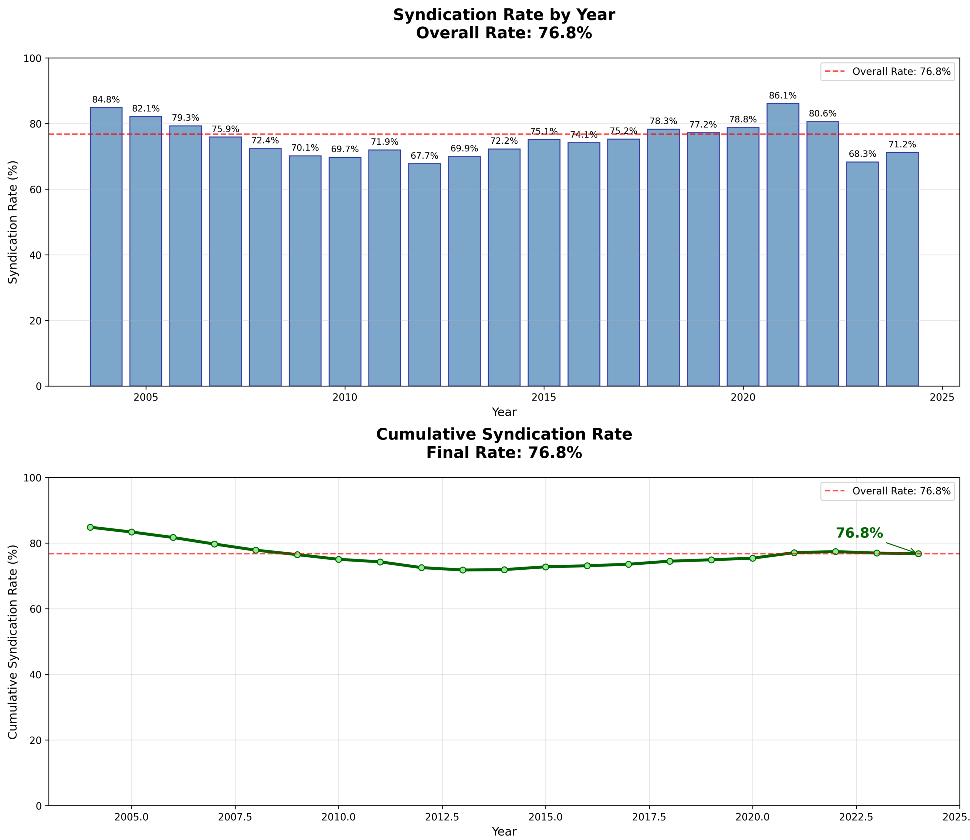


Figure 6: Syndication rate in the dataset. Top: annual share of syndicated rounds; bottom: cumulative share over time. The dashed line marks the overall rate.

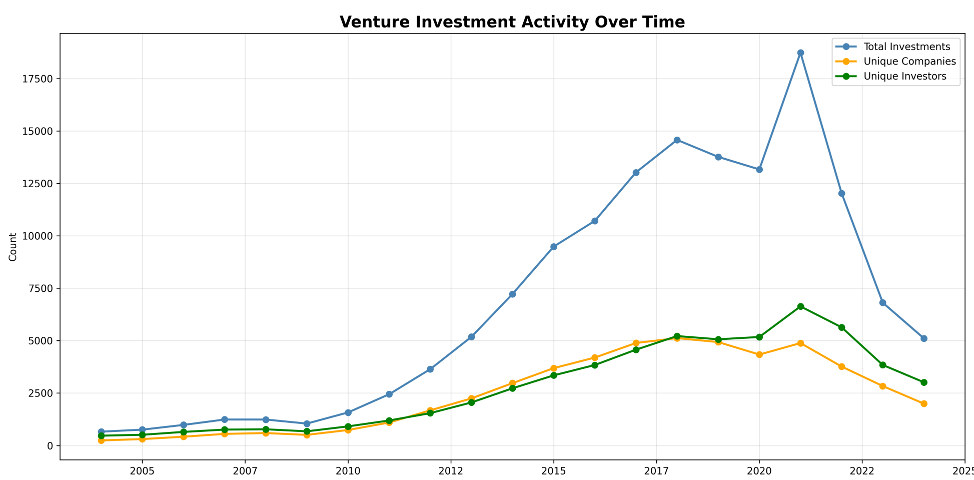


Figure 7: Total investments, unique companies, and unique investors by year.

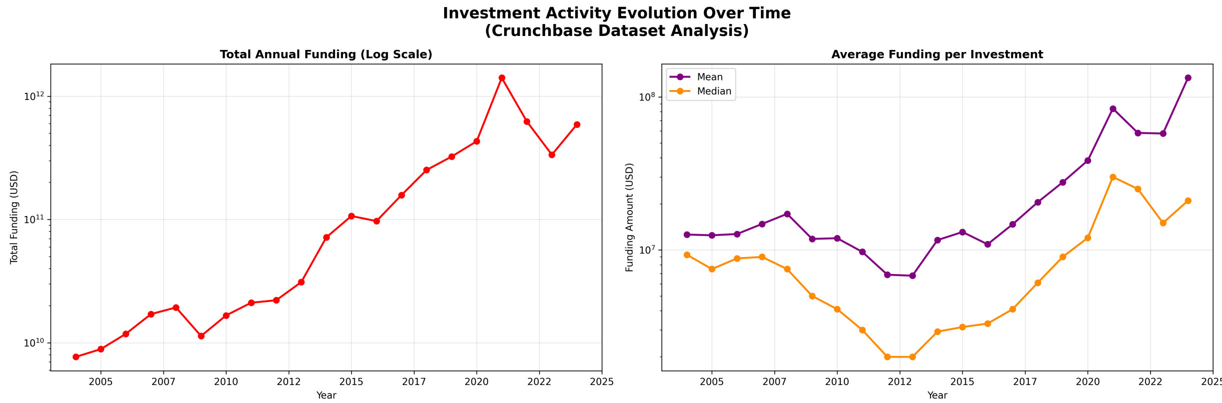


Figure 8: Temporal evolution of investment activity characteristics across the study period.

3.1.3 Investment Stages Evolution

Figures 9 and 10 illustrates the analysis of investment stages over time across the venture capital ecosystem

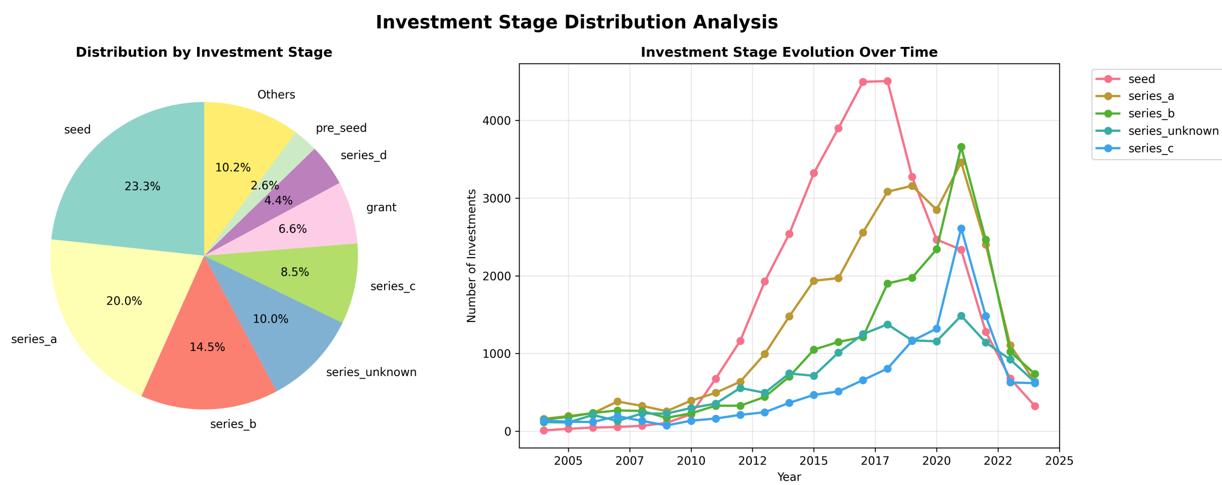


Figure 9: Distribution of investment stages across the venture capital dataset. The figure shows the prevalence of different funding stages, highlighting the concentration of activity in early-stage investments (seed and Series A) which together account for over 43% of all transactions.

The investment stage distribution demonstrates a clear early-stage focus within the venture capital ecosystem. Seed funding represents the largest single category at 23.3% of all investments (33,419 transactions), followed closely by Series A at 20.0% (28,703

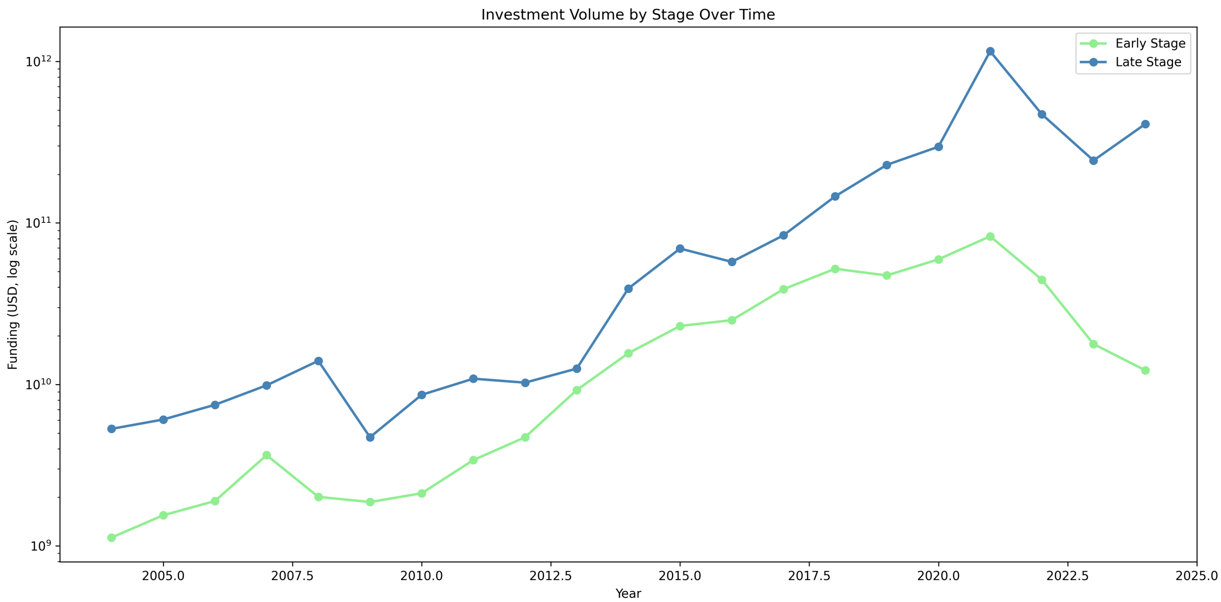


Figure 10: Stage composition over time (log scale for counts).

transactions). This concentration in early-stage funding reflects the fundamental role of venture capital in supporting nascent companies during their most critical development phases.

When aggregated by stage classification, early-stage investments (angel through Series A) comprise 46.7% of all transactions (66,978 investments), substantially exceeding late-stage investments (Series B and beyond) at 30.4% (43,635 investments). The remaining 22.9% consists of alternative funding mechanisms including grants, convertible notes, and specialized instruments.

However, temporal analysis also reveals a shift in capital allocation patterns. While early-stage investments dominate by transaction count, late-stage funding has achieved financial dominance in recent years. From 2004 to 2024, late-stage funding experienced exponential growth, escalating from \$5.31 billion in 2004 to \$409.52 billion in 2024 (a 77 times increase). In contrast, early-stage funding grew from \$1.13 billion to \$12.24 billion over the same period, representing an 11 times increase.

This divergence became particularly pronounced after 2013, when late-stage funding began substantially outpacing early-stage investments. By 2021, late-stage funding reached an unprecedented $1.16 trillion, passing the $82.74 billion in early-stage funding by a factor of 14.

Even during the market correction of 2022-2024, late-stage funding maintained its dominance, suggesting a structural trend toward growth-stage capital deployment.

This capital concentration in late-stage investments may indicate the venture capital industry's evolution toward supporting unicorn creation and pre-IPO growth, where individual transactions can exceed hundreds of millions of dollars. The trend suggests that while early-stage investments remain crucial for company formation, the majority of venture capital deployment now focuses on scaling proven business models.

3.1.4 Geographic Distribution

As expected, as our dataset query was built to get US companies, the geographic distribution of venture capital investments reveals US-concentration pattern on global scale (Figure 11). At US regional levels, Californian region (where Sylicon Valley is located) concentrates the invesment registers, as illustrated in Figure 12.

Globally, the United States accounts for 118,859 investments (82.9% of total activity), while international markets collectively represent 24,579 investments (17.1%). Among international markets, the United Kingdom leads with 3,210 investments (2.2%), followed by China with 1,855 investments (1.3%) and Canada with 1,743 investments (1.2%).

Within the United States, regional concentration patterns mirror the global trend. California dominates with 48,033 investments (40.4% of US total), representing nearly half of all American venture capital activity. New York follows as the second-largest market with 18,755 investments (15.8%), while Massachusetts ranks third with 7,946 investments

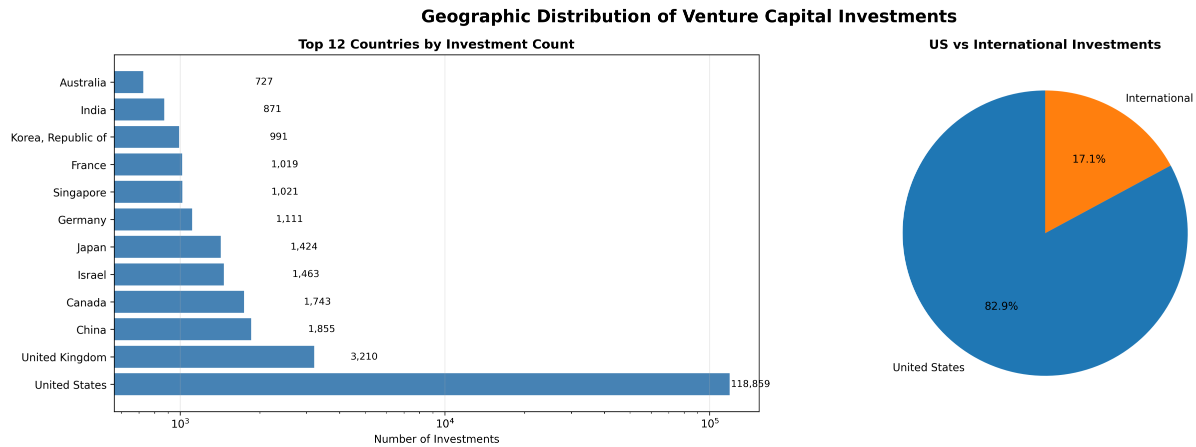


Figure 11: Geographic distribution of venture capital investments by country. The analysis shows the dominance of the United States VCs in venture capital activity on US (naturaly), accounting for 82.9% of all investment transactions, with the top 15 countries representing the vast majority of global venture capital deployment on US.

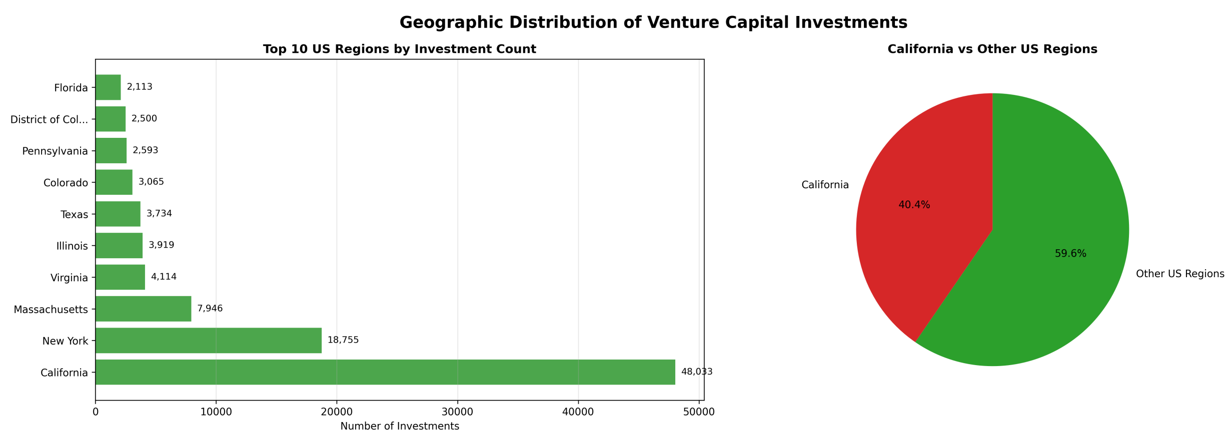


Figure 12: Regional distribution of venture capital investments within the United States. California emerges as the dominant hub with 40.4% of all US investments, followed by New York at 15.8%, demonstrating significant geographic clustering within the American venture capital ecosystem.

(6.7%). The top ten US regions account for approximately 81.5% of all domestic venture capital transactions.

These geographic patterns reflect the clustering effects of venture capital ecosystems around established technology and financial centers. The concentration in California, particularly in Silicon Valley, and New York's financial district demonstrates how proximity to talent, infrastructure, and capital sources influences investment distribution. Similar clustering patterns emerge internationally, with London, Beijing, and Toronto serving as regional venture capital hubs for funds coming to US.

3.1.5 Sectorial Distribution

The sectoral distribution analysis reveals significant concentration patterns in venture capital investment activity. As illustrated in Figure 13, health care dominates the investment landscape with 21,574 investments (15.0% of total activity), followed by financial services at 11,409 investments (8.0%) and data and analytics at 8,626 investments (6.0%).

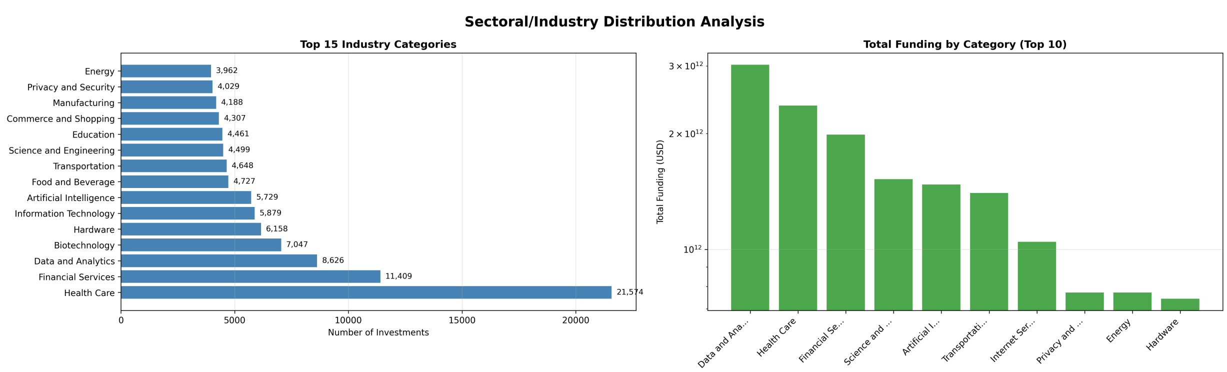


Figure 13: Sectoral distribution of venture capital investments across industry categories. The bar chart shows the concentration of investment activity, with health care as the dominant sector, followed by financial services and data analytics. The top 20 categories account for over 81% of all investment transactions.

The investment concentration demonstrates a highly uneven distribution across sectors. The top 5 categories capture 38.2% of total investments, while the top 10 categories account for 56.0% of all activity. This concentration extends to the top 20 categories, which represent 81.2% of total investments across only 42 unique industry categories.

Technology-related sectors represent a notable but minority portion of the ecosystem. The analysis identifies 6 technology categories (artificial intelligence, data and analytics, hardware, information technology, privacy and security, and internet services) that collectively account for 33,998 investments (23.7% of total activity). The remaining 109,440 investments (76.3%) occur in non-technology sectors, indicating that venture capital deployment extends well beyond traditional technology boundaries.

Figure 14 illustrates the temporal evolution of sectoral investment patterns, revealing how industry preferences have shifted over time.

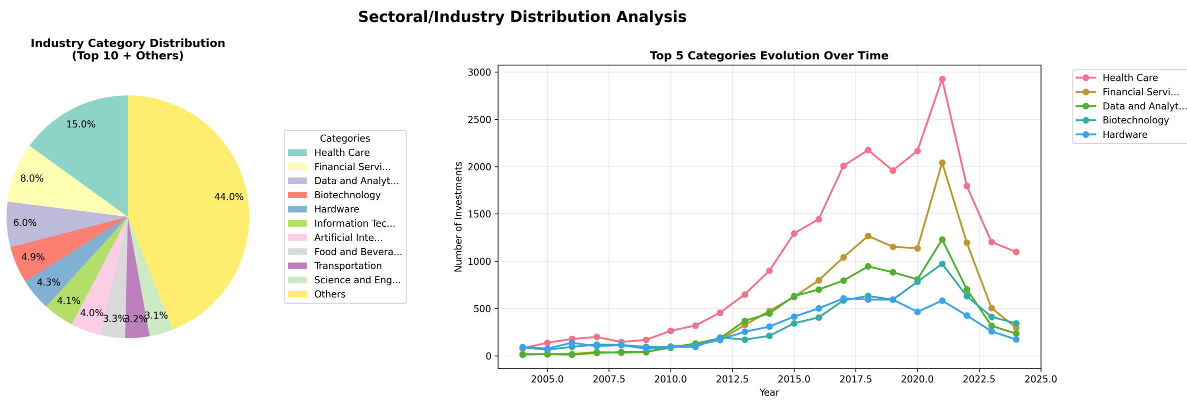


Figure 14: Left: category distribution (Top 10 + Others). Right: yearly evolution of the top five categories by number of investments.

The healthcare sector's dominance reflects the substantial capital requirements and long development cycles characteristic of medical innovation, which align well with venture capital investment strategies. Financial services' strong showing demonstrates the ongoing fintech revolution and the digitization of traditional financial institutions.

The presence of traditional sectors like food and beverage (4,727 investments, 3.3%), transportation (4,648 investments, 3.2%), and manufacturing (4,188 investments, 2.9%) among the top categories indicates that venture capital has expanded beyond its historical technology focus to encompass a broader range of innovation-driven industries.

Biotechnology's position as the fourth-largest category (7,047 investments, 4.9%) highlights the significant role of life sciences in the venture capital ecosystem, often requiring specialized knowledge and longer investment horizons compared to other sectors.

3.2 Communities Characterization

The division of venture capital firms into early-stage and late-stage investor groups results in 169,679 investment pairs comprising 3,666 unique startups.

Community detection using greedy modularity optimization identifies approximately 168 distinct communities (the number of communities oscillates between 167 and 175 across different trials), with the largest communities containing over 4000 investors each, followed by 1 community with almost 1000 agents, 4 communities with more than 100 agents, and then several smaller groups, as summarized in Table 1.

Initially, analysis focuses on communities with at least 150 nodes to ensure statistical power for nestedness analysis. This threshold yields 5 communities.

The largest three communities (0, 1 and 2) contain more than 12,000 investors combined, representing approximately 75% of all investors in the network. This concentration suggests a highly centralised structure within the venture-capital ecosystem, with most investment activity occurring within a small number of large communities. Such heavy-tailed community-size distributions are typical of social networks characterised by preferential attachment [6, 13].

Community boundaries are defined at the node level, meaning each investor belongs to exactly one community. However, edges (investment relationships) can span community boundaries when investors from different communities co-invest in the same startup.

Community ID	Number of Investors
0	4,248
1	4,089
2	3,959
3	979
4	188
5	155
6	137
7	122

Table 1: Sizes of the largest investor communities identified by greedy modularity.

To analyse investment patterns we classify each syndicated investment as either (1) intra-community, if all participating investors belong to the same community, or (2) cross-community, if investors from multiple communities participate together. Table 2 presents the resulting investment distribution across the three largest communities. Community 2 accounts for one-third of all investments despite being comparable in size to Communities 0 and 1, highlighting structural differences between communities.

Community	Co-investments Pairs	Relative Proportion
Community 0	32,164	19.4%
Community 1	17,301	10.4%
Community 2	55,863	33.6%
Cross-community	58,329	35.1%

Table 2: Distribution of co-investment pairs by community and cross-community share.

The analysis reveals important patterns in investment activity distribution. Community 2 accounts for the largest share of investments (33.6%), containing approximately 50% more investments than Community 0 and over three times more than Community 1. This concentration of investment activity suggests that structural features of Community 2 may facilitate higher transaction volumes. Notably, cross-community investments represent over one-third of all transactions, indicating substantial interconnectedness across community boundaries.

3.2.1 Degree Distribution

Before further examining community-level investment patterns in-depth, we analyze the degree distribution characteristics across the three largest investor communities. This analysis reveals that all communities exhibit power-law patterns typical of scale-free networks, with notable differences in magnitude and scale parameters.

Communities 0 and 2 demonstrate remarkably similar degree distribution magnitudes, while Community 1 exhibits consistently lower magnitude values across all degree ranges. This pattern becomes particularly evident when examining the degree distributions on a logarithmic scale, where Community 2 shows the highest volume of nodes across all degree ranges, suggesting greater overall connectivity within this community.

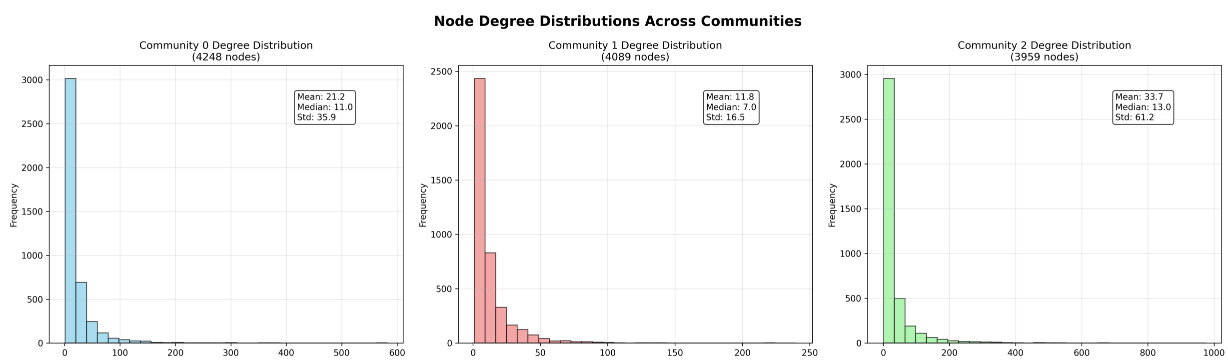


Figure 15: Node degree distributions for the three largest investor communities. The figure shows both the individual degree distributions and their overlap on a logarithmic scale. Communities 0 and 2 display similar heavy-tailed, power-law-like patterns, while Community 1 has consistently lower degree magnitudes. The log-scale overlay highlights that Community 2 maintains the highest node volume across all degree ranges, indicating greater overall connectivity.

Analysis of high-degree nodes (95th percentile) reveals distinct patterns across communities. Community 0 demonstrates a mixed composition of hub nodes, with early-stage investors like Techstars-seed (degree: 582) and 500 Global-seed (degree: 564) dominating the highest positions, alongside significant late-stage players such as Gaingels-series_b (degree: 509).

Community 1 exhibits a more balanced distribution with Intel Capital maintaining strong presence across multiple investment stages, while Community 2 shows remarkable concentration among early-stage Silicon Valley investors, with SV Angel-seed achieving the highest connectivity (degree: 974) followed by other prominent early-stage firms including Andreessen Horowitz and Khosla Ventures.

Table 3 presents the top high-degree nodes for each community, highlighting the structural differences in hub organization.

As per Figure 16, the relationship between degree centrality and investment activity demonstrates a positive correlation across all communities, with higher-degree nodes exhibiting greater investment frequency. This pattern suggests that network position, as measured by degree centrality, serves as a reliable predictor of investment activity levels within the venture capital ecosystem.

3.2.2 Geographic Distribution

Geographic analysis reveals distinct spatial clustering patterns across the three largest communities. Figure 17 illustrates the asymmetric geographic distributions between early-stage and late-stage investment networks.

Communities 0 and 2 exhibit similar geographic profiles with predominantly American investors, reflecting the dominance of U.S.-based venture capital (important to remember our dataset contains only American startups' investments, which include international investors).

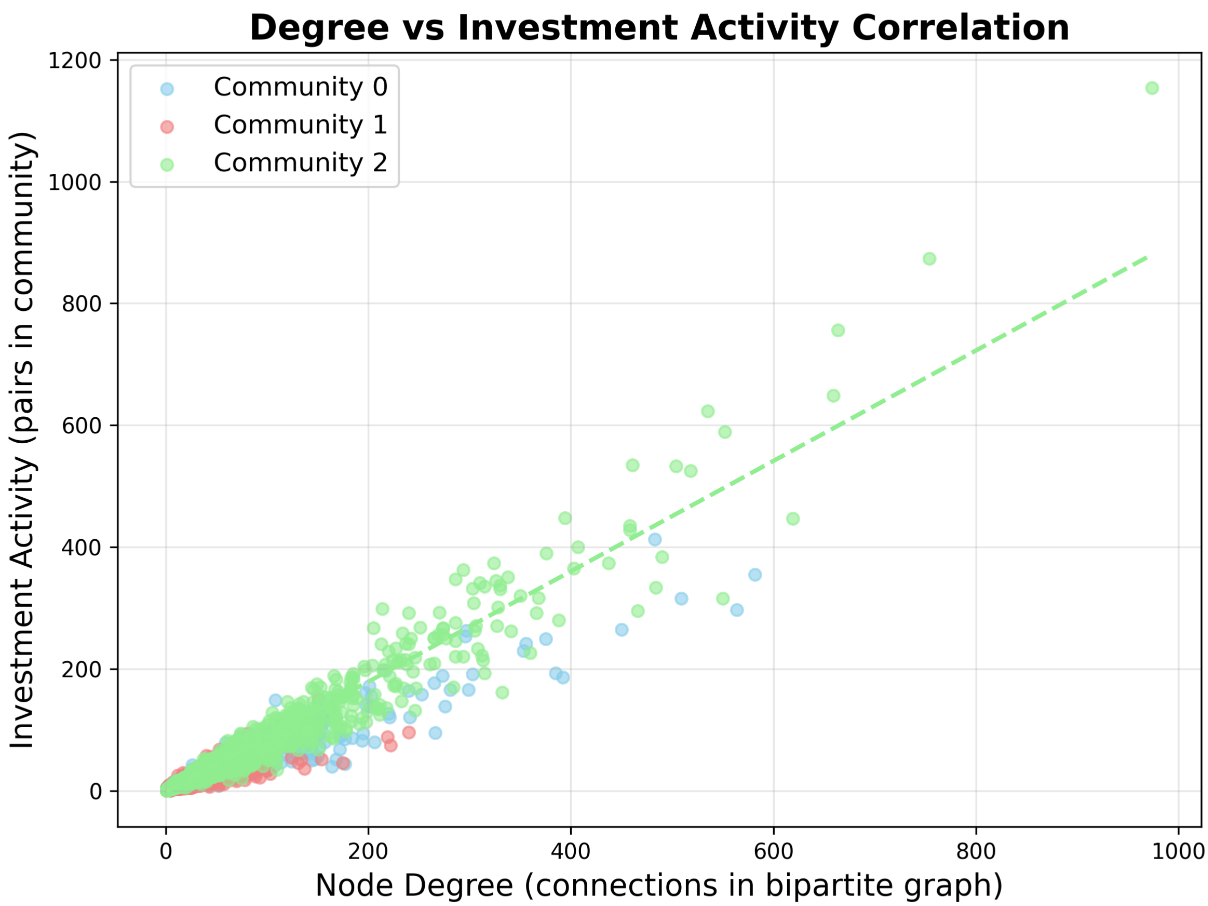
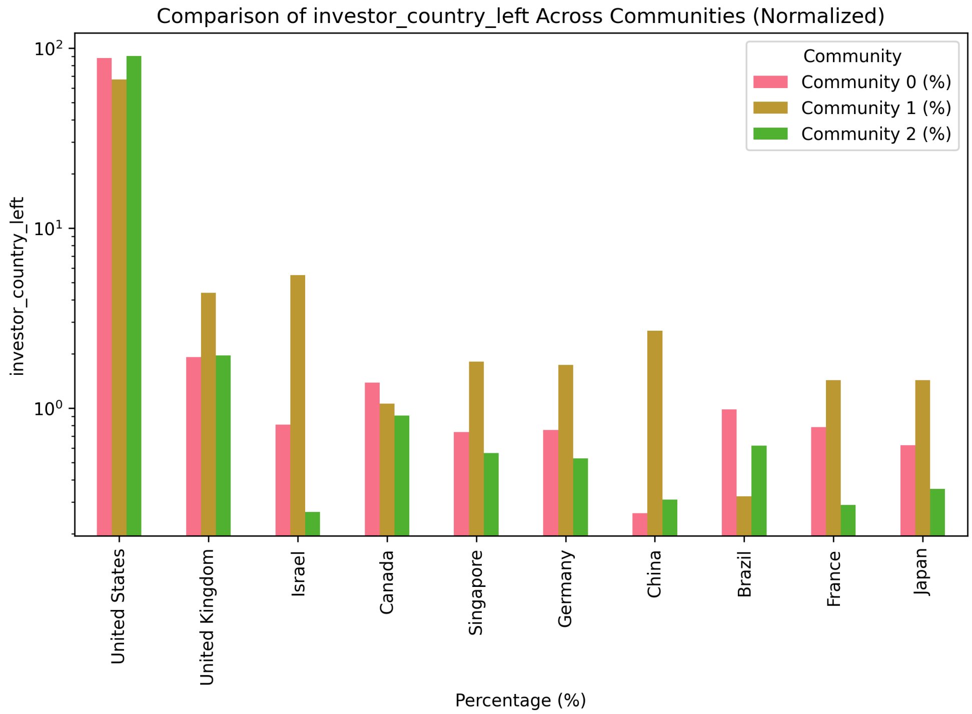
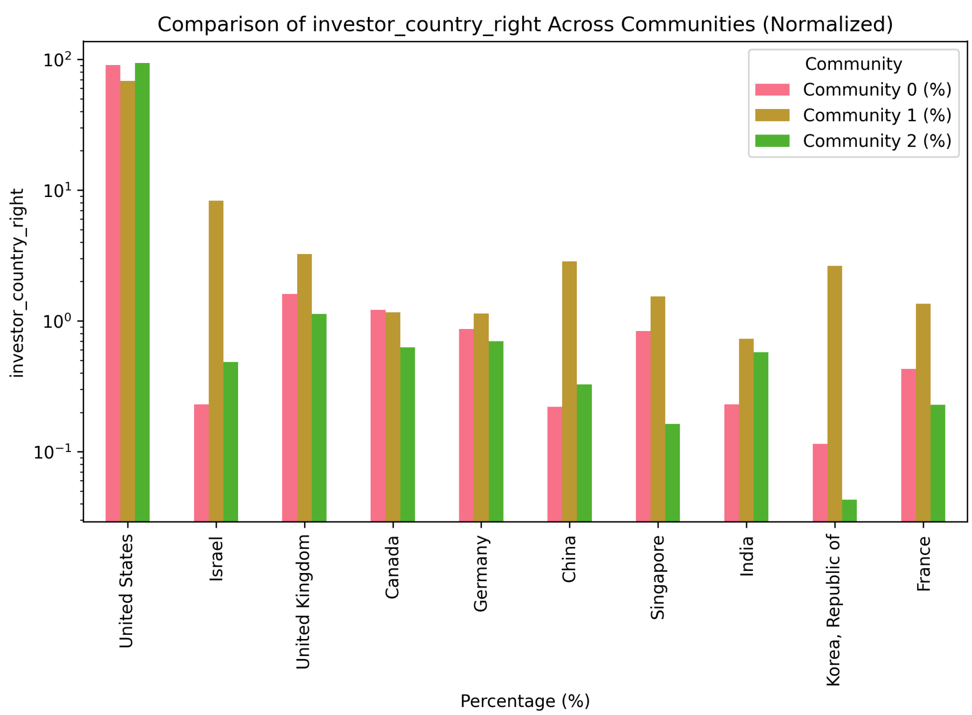


Figure 16: Relationship between node degree and investment activity for the three largest investor communities. The scatter plot demonstrates a positive correlation: higher-degree nodes tend to participate in more investment activities. This pattern is consistent across all communities, supporting the interpretation that network centrality is a strong predictor of investment frequency and influence within the venture capital ecosystem.

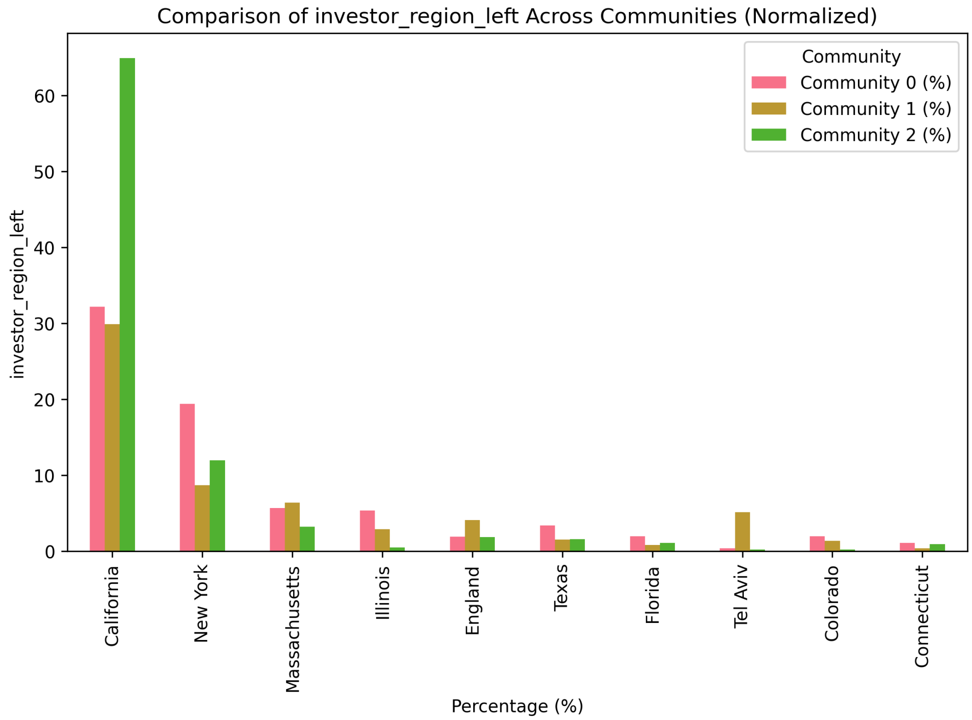


(a) Late-stage investors by country (log scale)

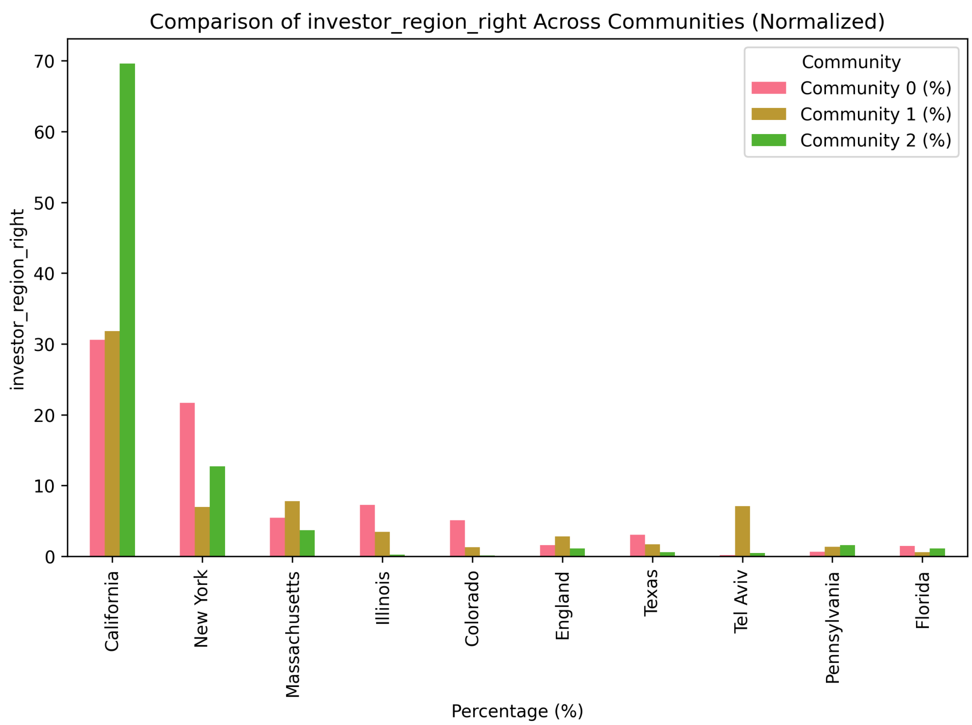


(b) Early-stage investors by country (log scale)

Figure 17: Geographic distribution across countries for late-stage (top) and early-stage (bottom) investors in the three largest communities.



(a) Late-stage investors by region



(b) Early-stage investors by region

Figure 18: Geographic distribution across regions (worldwide) for late-stage (top) and early-stage (bottom) investors in the three largest communities.

Community	Investor	Degree	Type
0	Techstars-seed	582	Early-stage
	500 Global-seed	564	Early-stage
	Gaingels-series_b	509	Late-stage
	Greycroft-series_a	483	Early-stage
	Bossa Invest-series_b	450	Late-stage
1	Intel Capital-series_b	240	Late-stage
	Norwest Venture Partners-series_a	222	Early-stage
	Canaan Partners-series_a	219	Early-stage
	Intel Capital-series_c	175	Late-stage
	SOSV-series_b	163	Late-stage
2	SV Angel-seed	974	Early-stage
	SV Angel-series_a	754	Early-stage
	Andreessen Horowitz-series_a	664	Early-stage
	Khosla Ventures-series_a	659	Early-stage
	New Enterprise Associates-series_a	619	Early-stage

Table 3: Top 5 high-degree investors (95th percentile) by community and stage.

However, regional analysis within the United States reveals a striking pattern: Community 2 demonstrates exceptional concentration in California, particularly Silicon Valley, with approximately 50% more California-based investors than either Community 0 or Community 1 for both late-stage and early-stage investor categories.

This Silicon Valley concentration in Community 2 is particularly notable given the region's status as the world's premier innovation ecosystem. The dominance of California investors in this community, which also exhibits the highest transaction volumes, suggests potential advantages conferred by geographic clustering within innovation hubs.

In contrast, Community 1 demonstrates significantly greater international diversification, with substantial representation from Israel, the United Kingdom, China, South Korea, Singapore, and France. This international composition in Community 1 may reflect different risk tolerance profiles, regulatory environments, or access to cross-border deal flow compared to the more domestically concentrated communities.

3.2.3 Funding Characteristics

Analysis of funding patterns reveals that Community 2 exhibits substantially higher funding frequency and larger investment amounts compared to the other communities, as demonstrated in Figure 19.

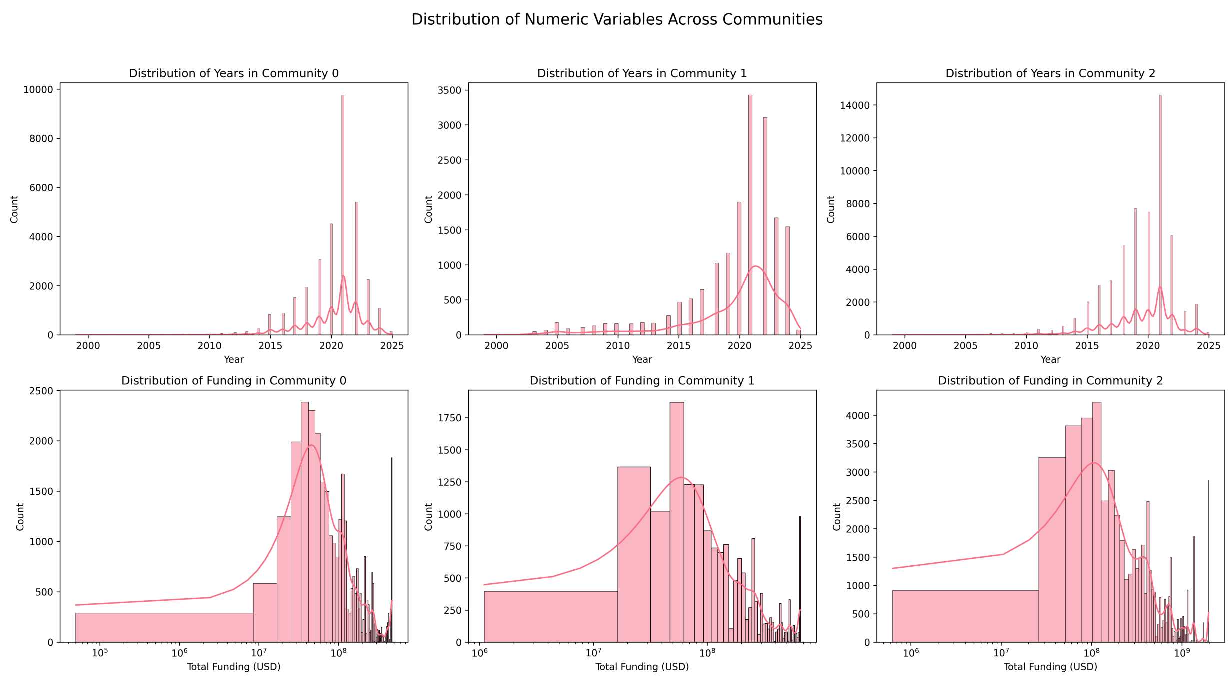


Figure 19: Distributions across communities. Top row: histograms of deal years. Bottom row: log-scale histograms of total funding per round with KDE overlays.

Furthermore, communities exhibit concentrated capital deployment patterns, with higher-degree investors participating in larger funding rounds while maintaining broader portfolio diversification. This suggests that certain organizational structures may create more efficient capital allocation mechanisms compared to other network configurations.

Despite superficial similarities between Communities 0 and 2 in terms of size and geographic concentration within the United States, Community 2 appears to confer distinctive advantages in organizational efficiency. While both communities share comparable scales and American investor bases, Community 2's organizational structure enables it to achieve substantially higher transaction volumes (33.6% vs. 19.4% of total investments) and more comprehensive funding coverage across all investment stages.

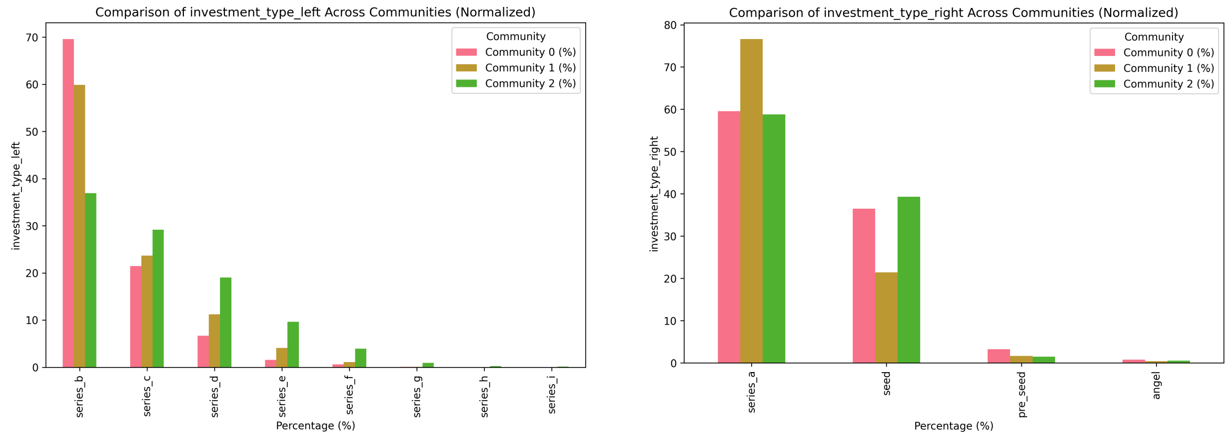
This pattern suggests that specific network topologies, rather than community size or geographic distribution alone, may be critical determinants of investment ecosystem efficiency.

The interpretation that network topology influences capital deployment efficiency is motivated by theory on preferential attachment and small-world networks [6, 15] and by ecological studies linking nestedness to resource flow efficiency [23, 26].

3.2.4 Investment Stage Preferences

The investment stage distributions reveal systematic specialization patterns across communities that align with their geographic profiles and transaction volumes. Figure 20 demonstrates the distribution patterns of investment types within the bipartite network structure.

Late-stage investment patterns: Community 2 dominates Series C and later funding rounds, demonstrating its role in growth-stage capital deployment. Community 1, despite its smaller transaction volume, shows strong representation in Series B and later stages, with participation rates exceeding Community 0 in Series C and beyond. Community 0 exhibits particular strength in Series B rounds while maintaining lower participation in



(a) Late-stage investment types distribution

(b) Early-stage investment types distribution

Figure 20: Investment stage distribution across the three largest communities. The distribution patterns reveal stage-specific specialization within investor communities.

later stages compared to Community 2.

Early-stage investment patterns: Community 2 shows prominence in seed-stage investments while maintaining comparable levels to Community 0 in Series A funding. Both Communities 0 and 2 participate actively in angel and seed rounds, though Community 0 shows relatively higher pre-seed activity. Community 1 demonstrates concentrated focus on Series A investments, aligning with its international profile and suggesting specialization in cross-border early-growth funding.

These stage-specific patterns suggest that Community 2's organizational structure facilitates participation across the entire funding spectrum, from seed to late-stage rounds, potentially enabling more comprehensive support for portfolio companies throughout their development lifecycle.

3.2.5 Sectoral Focus

The sectoral analysis reveals distinct specialization patterns that align with each community's structural and geographic characteristics. Figure 21 illustrates the distribution of investment focus across technology sectors, demonstrating how different communities exhibit varying degrees of sectoral concentration.

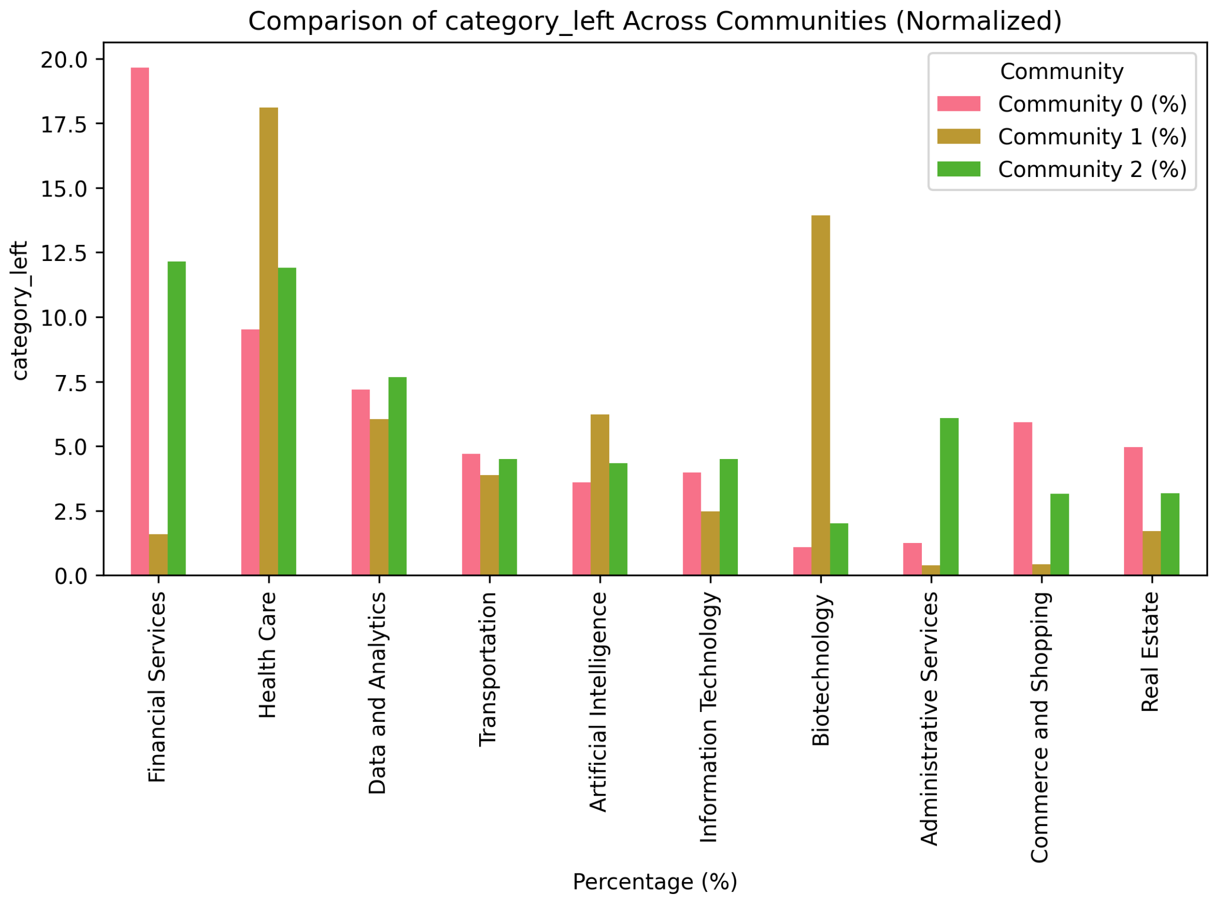


Figure 21: Sectoral distribution across the three largest investor communities. The analysis reveals differential industry focus patterns, with certain communities demonstrating concentrated investment strategies in specific technology sectors while others maintain broader sectoral diversification.

Community 0: Concentrates in real estate, commerce and shopping, and financial services, while maintaining standard representation in information technology and artificial intelligence. Notably absent from administrative services and biotechnology, suggesting specialized expertise in consumer-facing and financial technology sectors.

Community 1: Specializes significantly in biotechnology and healthcare, with enhanced artificial intelligence focus compared to other communities. Shows reduced interest in real estate, commerce and shopping, and administrative services. The biotechnology concentration aligns with its international composition, potentially reflecting access to global biotech innovation hubs.

Community 2: This community exhibits strong representation in administrative services while maintaining comparable levels to Community 0 in information technology and artificial intelligence. Despite its larger transaction volume, it shows lower concentration in real estate and financial services than Community 0, suggesting that its organizational structure facilitates broader sectoral participation rather than concentrated specialization. This sectoral breadth, combined with the community's Silicon Valley concentration, indicates potential advantages of diversified investment strategies within innovation-rich geographic clusters.

Community 0 serves as an effective structural baseline for comparison with Community 2, given their similar geographic profiles and certain sectoral overlaps, while differing significantly in network organization and transaction volumes. The systematic differences observed between these structurally similar communities highlight the potential impact of organizational structures on investment behavior and capital deployment efficiency.

3.3 Overall Nestedness Findings

Nestedness analysis across investor communities reveals heterogeneous structural patterns. Among the 8 communities examined, one exhibits significantly high nestedness ($p < 0.01$) relative to degree-preserving null models generated through the Curveball algorithm.

To provide a concise overview of community characteristics in that regard, Table 4 summarises key statistics for the three largest communities. In addition to the number of investors and co-investment pairs, it reports the share of intra-community investments and the observed nestedness and p-value from degree-preserving null models. Only Community 2 (msotly composed by Silicon Valley investors) exhibits nestedness significantly higher than expected under random mixing.

Community	Investors	Syndication pairs	Investment share	Observed NODF	P-value
0	4,248	32,164	19.4%	0.028	1
1	4,089	17,301	10.4%	0.015	1
2	3,959	55,863	33.6%	0.088	<0.001

Table 4: Three largest communities: investors, co-investment pairs, share of intra-community investments, observed NODF, and p-value (degree-preserving null model) for the hypothesis of nestedness higher than null expectation.

Nevertheless, Figure 22 presents the comparison between observed and null model nestedness scores, where each plot represents a distinct community's null model NODF distribution accompanied by its observed NODF value.

A broader analysis examining nestedness patterns across all analyzed communities reveals a relationship between community size and nestedness significance, as illustrated in Figure 23. This analysis examines 8 communities ranging from 122 to 4,248 investors, providing insights into how community structure varies across different organizational scales.

The analysis reveals that high nestedness significance does not follow a simple relationship with community size. While smaller communities (122-188 investors) tend to exhibit higher absolute nestedness scores (0.045-0.099), these values are statistically significant low when compared to appropriate null models. Medium-sized communities (979-4,248 investors) generally show lower absolute nestedness scores (0.011-0.028), with most falling below low nestedness significance thresholds.

Notably, Community 2 stands out as the only community achieving statistically significant nestedness despite having a moderate absolute score (0.088) and being neither the

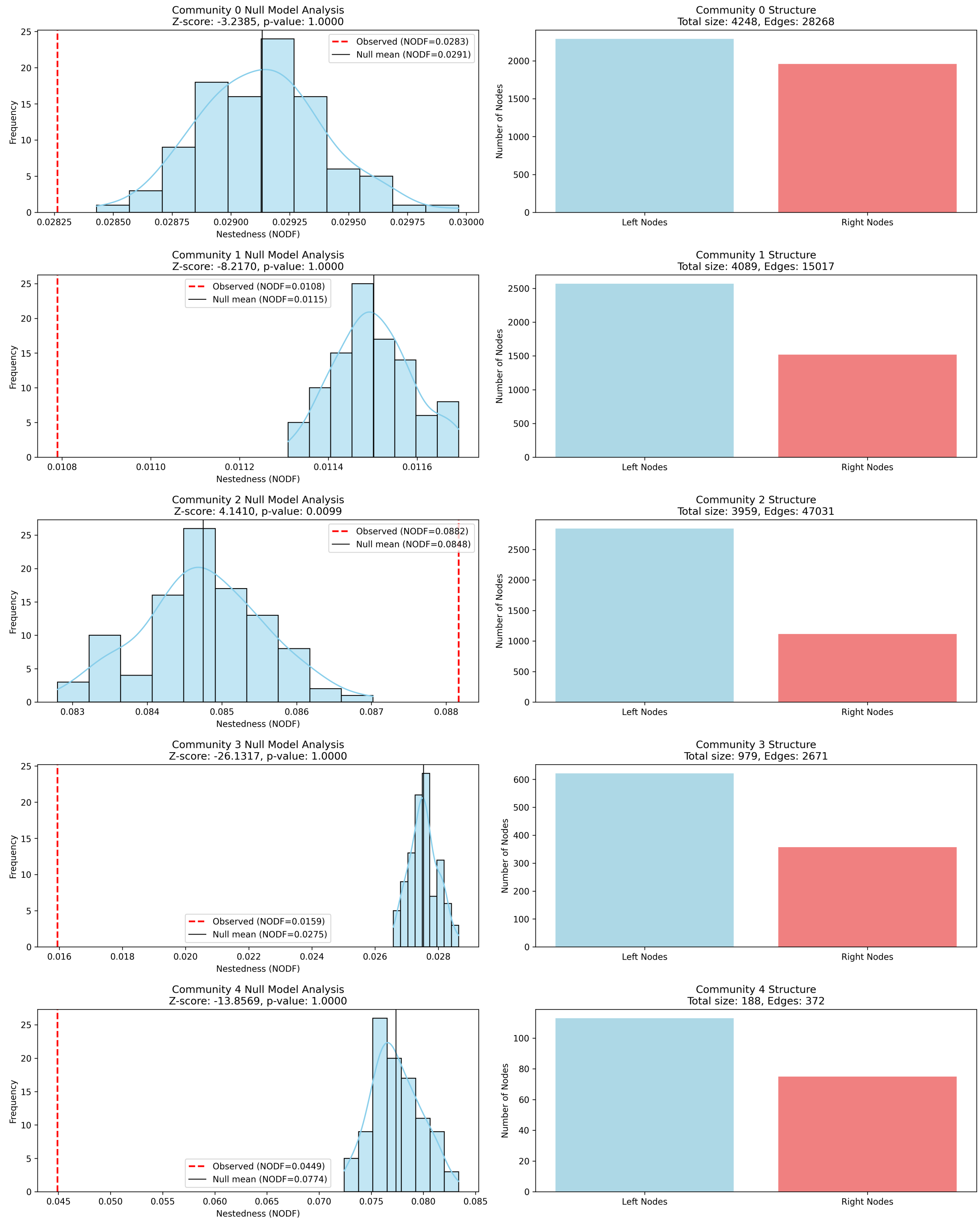


Figure 22: For each large community (rows): left, null-model NODF distribution (blue) with observed value (red); right, network size split into late-stage (left nodes) and early-stage (right nodes). Reported Z-scores and p-values summarize significance.

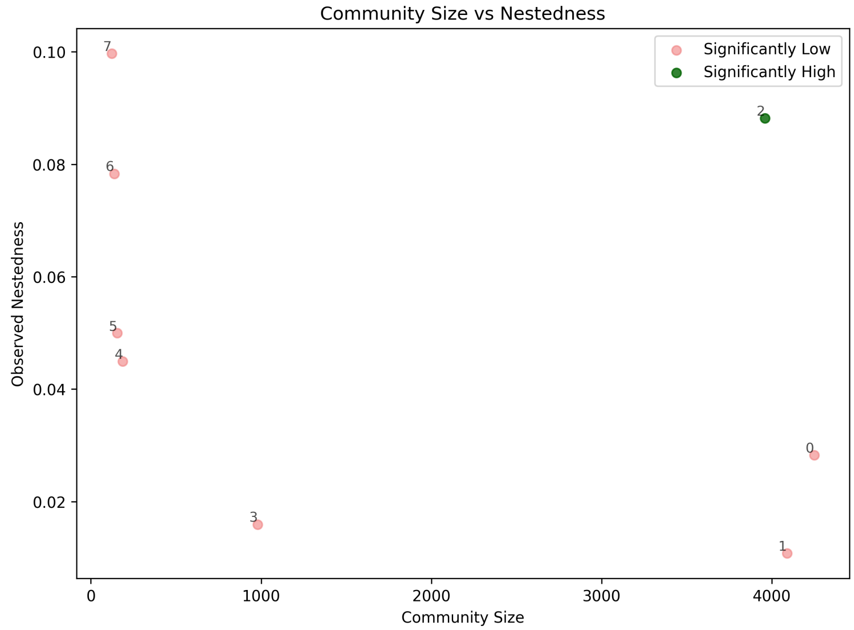


Figure 23: Relationship between community size and nestedness significance across all analyzed investor communities. The figure shows observed nestedness scores (y-axis) against community size (x-axis), with markers indicating statistical significance. Only Community 2 exhibits significantly high nestedness, while all other communities show significant low patterns relative to their respective null models.

largest nor smallest community analyzed. This pattern suggests that nestedness emergence depends on specific structural characteristics rather than simple community scale effects.

Community 2 demonstrates the most pronounced nestedness, exhibiting an NODF score of 0.088 with statistical significance of p = 0.001. This indicates a non-random behavior where less-connected investors maintain co-investment relationships with subsets of partners associated with highly-connected investors, creating a hierarchical investment pattern.

Additionally, Community 2 exhibits an asymmetric composition with a pronounced ratio favoring late-stage investors over early-stage investors. This imbalance may contribute to the nested structure [5] by creating hierarchical dependencies between investor types.

The following sections provide detailed characterization of the 3 most relevant communities in terms of number of investors, while analyzing in-depth Community 2 (where nestedness was observed) through comparison with Communities 0 and 1, which serve as contrasting examples of similar-sized but differently structured investor networks.

3.4 Evolution of Nestedness in Silicon Valley Community

To understand the emergence and development of nested structure, we conducted a temporal analysis of Community 2's nestedness evolution using cumulative window. This analysis reveals how the significantly high nested structure observed in the static analysis developed over time and when it became statistically distinguishable from random network configurations.

The temporal analysis spans 18 years (2007-2024) with sufficient data for nestedness calculation. Early periods (2004-2006) contained insufficient investment activity for meaningful analysis. The evolution demonstrates a progressive development pattern: an early period with significantly low nestedness (2007-2018), followed by the achievement of statistical significance in 2019, and a sustained period with significantly high nestedness (2019-2024).

During the period with significantly low nestedness (2007-2018), observed nestedness scores ranged from 0.15 to 0.38, consistently failing to exceed null model expectations. Z-scores remained predominantly negative, indicating that the observed network structure was less nested than expected under random configuration preserving the degree sequence. However, careful examination of the data reveals that nestedness values began showing gradual increases after 2012, suggesting the beginning of a progressive structural transformation.

Statistical significance was first achieved in 2019, marking when Community 2's nested structure became significantly distinguishable from random expectations (Z-score: 2.91, p-value < 0.001). This milestone coincided with substantial network growth, reaching 2,364 total nodes and 33,805 edges.

Notably, the achievement of statistical significance in 2019 corresponds to the network reaching a connectance threshold of 0.0264, suggesting that nestedness significance may be related to specific density conditions within large-scale investment networks, following a progressive build-up of hierarchical structures that began years earlier.

The sustained significant period (2019-2024) demonstrates consistent statistical significance with progressively strengthening Z-scores, reaching a maximum of 4.76 in 2022.

Interestingly, while absolute nestedness scores decreased from 0.38 in early periods to 0.088 in 2024, the statistical significance increased dramatically. Table 5 presents key statistics from the temporal evolution analysis.

This apparent paradox reflects the fundamental principle that nestedness significance depends on comparison with appropriate null models rather than absolute magnitude [5].

Period	**Years**	**Mean NODF**	**Mean Z-score**
Low Nestedness (2007-2018)	12	0.2547 ± 0.1158	-1.4442 ± 1.1108
Transition (2019)	1	0.1177	2.9125
High Nestedness (2019-2024)	6	0.1012 ± 0.0121	3.8950 ± 0.7182

Table 5: Community 2 nestedness over time: mean NODF and mean Z-scores.

The analysis reveals several notable patterns in network structure evolution. Late-stage investor participation increased consistently over time, while early-stage investor numbers showed initial growth followed by stabilization after 2020. This asymmetric growth pattern contributed to the development of the nested structure by creating increasingly hierarchical relationships between investor types.

Connectance exhibited a systematic decline from 0.195 in 2007 to 0.015 in 2024, reflecting the network's evolution toward sparser but more strategically organized connections.

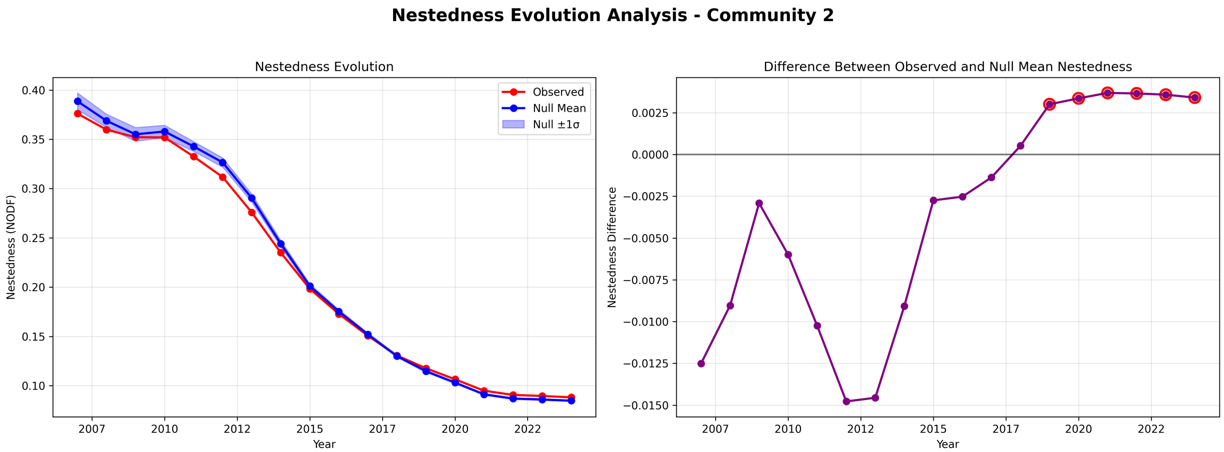


Figure 24: Temporal evolution of nestedness in Community 2. The left subplot shows the observed nestedness (red) and null model mean (blue) over time, with shaded areas representing the null model standard deviation. The right subplot displays the evolution of network size, with total nodes (green) and total edges (purple) over the same period. The figure highlights the phase transition in 2019, where observed nestedness becomes significantly higher than null model expectations and the network grows rapidly.

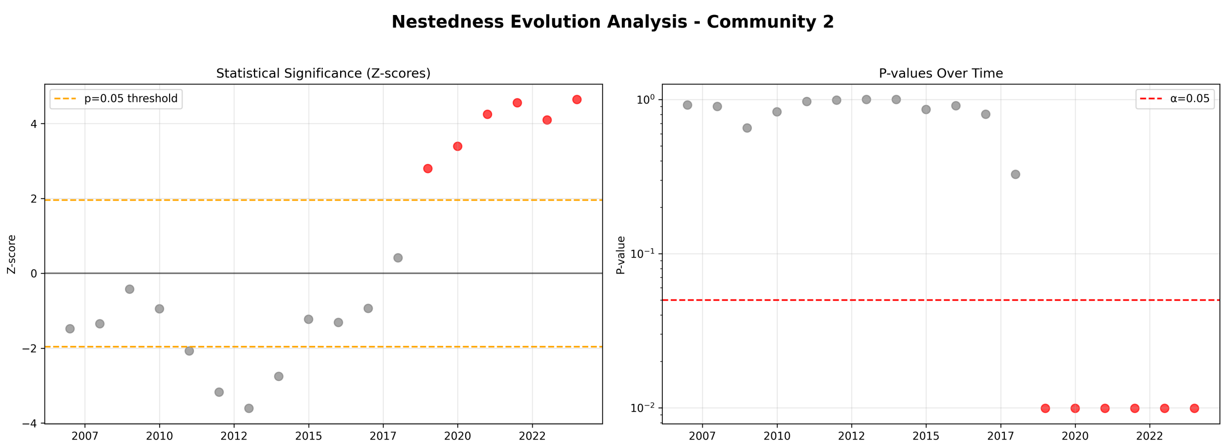


Figure 25: Connectance and statistical significance evolution in Community 2. The left subplot shows the decreasing trend in connectance (network density) over time, while the right subplot presents the evolution of p-values on a logarithmic scale, indicating the achievement of significantly high nestedness statistical significance in 2019. The figure demonstrates how the network becomes sparser yet more hierarchically organized, with nestedness statistical significance achieved as the network reaches critical size and density thresholds.

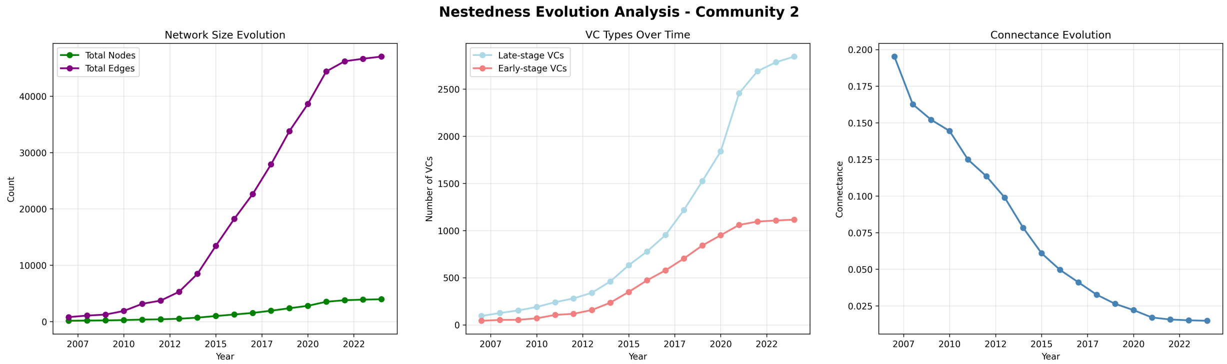


Figure 26: Summary for Community 2 (2007–2024). Left: total nodes and edges; middle: counts of late- vs early-stage investors; right: connectance.

Despite this decreased density, the achievement of statistical significance suggests that the remaining connections became increasingly hierarchically organized, with less-connected investors maintaining relationships with subsets of the partners of highly-connected investors.

Detailed analysis of the significant periods (2019-2024) reveals consistent patterns in network organization. Each significant year demonstrates similar structural characteristics: large networks (>2,300 nodes), substantial edge counts (>33,000), low connectance (<0.027), and strong statistical significance (Z-scores >2.9). This consistency suggests that the nested structure represents a stable organizational state that persists once established.

The temporal analysis provides evidence that nestedness in venture capital networks develops through a progressive evolution process rather than sudden emergence. The gradual increase beginning after 2012, followed by the achievement of statistical significance in 2019 and sustained high nestedness thereafter, suggests that hierarchical organization builds up over time before crossing significance thresholds.

This pattern aligns with theoretical frameworks from complex network theory indicating that certain topological properties can develop progressively as networks reach critical size or density parameters [5]. The underlying mechanisms driving this progressive development remain unexplored in this work, representing an important avenue for future research.

The period of nestedness statistical significance achievement (2019-2024) corresponds with an increase in late-stage investor participation and a relative decrease in early-stage investor numbers within the community. This asymmetric evolution may contribute to the hierarchical structure by creating conditions where early-stage investors increasingly depend on relationships with a subset of the partners associated with highly-connected late-stage investors.

The evolution of nestedness versus null models between 2019 and 2024 is plotted in Figure 24. It shows that observed nestedness scores increasingly exceed null expectations during the high-nestedness period.

However, whether this temporal correlation reflects causal mechanisms or represents coincidental market dynamics requires further investigation with appropriate theoretical frameworks [5]. Future work could combine network analysis with econometric models to explore causality.

4 Discussion and Implications

As outlined in the Results section, the present work first presents a detailed characterization of investor communities and then examines nestedness within those communities; the discussion below follows that ordering and builds directly on the community-level descriptions provided earlier.

The discovery, documented in the Results section, of significantly high nestedness in the late-early stage VC community centred on the Silicon Valley (SV) provides new insights into investor behavior and startup access to capital.

First, it adds a significant geographic dimension to these findings, suggesting that geographic clustering may facilitate the emergence of nested hierarchical investment structures.

Second, the concentration of significantly high nested structure specifically within Silicon Valley, given its worldwide relevance, suggests that nested investment hierarchies may systematically influence funding accessibility for entrepreneurs.

Furthermore, the comparative analysis between Communities 0 and 2 (SV) reveals that high nestedness functions as an organizational catalyst that transforms otherwise similar investor communities. It indicates that SV's nested structure enables substantially higher transaction volumes and more funding coverage across all investment stages compared to Community 0, despite comparable sizes and geographic concentration within the United States.

This suggests that network topology, rather than community size or geographic distribution alone, may be determinant of investment ecosystem efficiency. The hierarchical organization in SV appears to create more efficient capital allocation mechanisms, with higher-degree investors participating in larger funding rounds while maintaining broader portfolio diversification compared to randomly structured networks.

4.1 Degree Distribution Patterns and Hub Organization

The analysis of degree distributions across communities reveals fundamental organisational differences that complement the nestedness findings. All three communities exhibit heavy-tailed degree distributions characteristic of scale-free networks [6, 13], but with distinct magnitude and structural parameters that reflect different organisational strategies.

The similarity in degree distribution magnitudes between Communities 0 and 2, contrasted with Community 1's consistently lower values, suggests that network scale alone does not determine investment efficiency. SV's superior performance in investment volumes and nested organization occurs despite degree distribution patterns nearly identical to Community 0.

This finding reinforces that the specific arrangement of connections, rather than their quantity or distribution, drives the observed efficiency advantages.

Analysis of high-degree nodes (network hubs) reveals distinct organizational philosophies across communities. SV's hub structure demonstrates exceptional concentration among early-stage Silicon Valley investors, with SV Angel achieving remarkable connectivity across both seed and Series A stages.

This pattern contrasts with Community 0's more diversified hub structure and Community 1's balanced early-late stage distribution.

The concentration of hubs within the early-stage investment category in SV may facilitate the nested structure by creating clear hierarchical pathways from early-stage to late-stage investment relationships.

The positive correlation between degree centrality and investment activity across all communities validates network position as a predictor of investor influence. However, the strength of this relationship appears amplified within the nested SV, suggesting that hierarchical organization may enhance the efficiency of high-degree investors in deploying capital and identifying investment opportunities.

4.2 Temporal Dynamics and Progressive Development

The temporal evolution analysis of SV provides unprecedented insight into how highly nested structures emerge within VC-VC investment networks. The identification of a progressive development pattern with statistical significance achieved in 2019, rather than abrupt nestedness emergence, challenges assumptions about evolutionary network organization and suggests progressive accumulation mechanisms that cross significance thresholds.

The three-phase evolution pattern (period with significantly low nestedness between 2007-2012, transition between 2013-2019, and sustained high nestedness during 2019-2024) indicates that nested organization in venture capital networks develops progressively before crossing significance thresholds. This progressive development followed by threshold crossing aligns with theoretical predictions from complex network theory regarding gradual transitions in topological properties that become statistically detectable at critical points [5].

The apparent paradox of decreasing absolute nestedness scores (from 0.38 to 0.088) coinciding with increasing statistical significance reflects sophisticated changes in network organization. As the network grew substantially larger, maintaining even modest levels of hierarchical organization became increasingly difficult under random formation processes, making the observed nested patterns more statistically remarkable.

The correspondence between the emergence of high nestedness and specific connectance thresholds (approximately 0.026) provides practical insights for ecosystem development.

This threshold may represent a critical density where hierarchical organization becomes sustainable within large-scale investment networks, offering guidance for policy interventions aimed at fostering similar organizational efficiency in other innovation ecosystems.

The asymmetric evolution of investor types (increasing late-stage participation coupled with stabilizing early-stage numbers) may have facilitated the progressive development of nested structure by creating conditions favoring hierarchical relationships over time.

This pattern suggests that the development of nested organization may require specific demographic conditions within investor communities, rather than simply network growth or density changes. As mentioned in previous sections, the underlying mechanisms driving this progressive structural evolution remain unexplored in this work, representing an important avenue for future research into the causal factors behind hierarchical organization in innovation networks [5, 10, 33].

4.3 Network Robustness and Resilience

The presence of nested structures challenge assumptions of random mixing in venture-capital markets, suggesting that certain investors function as "gatekeepers" who control access to broader investment networks. This finding aligns with social-network theories about structural holes and brokerage positions [11, 13]. Thus, following insights from nestedness research in complex networks [5], the hierarchical organization observed in SV may confer distinct robustness properties to the venture capital ecosystem.

In mutualistic networks, high nestedness typically enhances stability against random node removal but creates vulnerability to targeted elimination of highly connected nodes. Applied to venture capital, this suggests that highly nested investor communities may be resilient to random investor departures, they could be particularly vulnerable to the exit of key hub investors.

The concept of "mutualistic trade-offs" from ecological network theory provides a framework for understanding these dynamics. In highly nested venture capital communities, less-connected investors maintain relationships with subsets of the partners associated with highly-connected investors, creating dependencies that could influence network stability.

Future research should investigate whether less-connected venture capital firms exhibit higher exit probabilities, which would support the hypothesis that high nestedness creates certain hierarchical fragility/robustness patterns.

4.4 Policy Implications and Future Research

Our findings have several implications for policymakers, investors and entrepreneurs. First, the emergence of nested, hub-dominated communities suggests that access to funding may depend not only on firm quality but also on network position.

Policymakers seeking to broaden access to venture capital could support initiatives that connect peripheral investors and entrepreneurs to central hubs or encourage the formation of new hubs in underserved regions.

Second, the strong association between geographic clustering and nestedness indicates that place-based policies, such as investments in innovation districts and infrastructure, may facilitate the development of efficient capital allocation networks.

However, nested organisation also entails vulnerabilities. Highly nested communities are resilient to random departures but fragile to the exit of key hub investors. Regulators and investors should monitor concentration risk and consider mechanisms that encourage diversification of relationships. From an entrepreneurial perspective, understanding network topology can help founders identify strategic co-investment partnerships that enhance visibility and funding opportunities.

5 Conclusion

This work applied tools from Network Science, economic sociology and ecology to analyse U.S. venture-capital syndication patterns. By constructing a bipartite co-investment network and applying modularity-based community detection and nestedness analysis, we uncovered substantial heterogeneity in organisational patterns across investor communities.

A community centred in Silicon Valley displayed significantly high nestedness: less-connected investors co-invested almost exclusively with subsets of the partners of highly connected investors. In contrast, similarly sized communities lacked hierarchical organisation and exhibited lower transaction volumes.

Temporal analysis revealed that nestedness follows a progressive development pattern, with values beginning to increase gradually after 2012 and achieving statistical significance in 2019, with this hierarchical structure persisting thereafter. This finding suggests that network topology can undergo progressive structural changes that accumulate over time before crossing statistical significance thresholds, echoing results from ecological and complex-systems research [5, 7].

The progressive emergence of nestedness coincided with increasing participation by late-stage investors and stabilisation of early-stage participation, pointing to demographic conditions that may favour hierarchical organisation. However, the underlying mechanisms driving this temporal evolution remain unexplored in this work, representing an important opportunity for future research into the causal factors behind progressive hierarchical organization in innovation networks [1, 2, 5].

These results underscore three key contributions. First, they demonstrate that network topology (particularly nested hierarchy) matters more than community size alone for understanding investment efficiency and access to capital. Second, they bridge insights from economic sociology and ecology, showing that venture-capital communities can resemble mutualistic ecosystems (e.g. pollinator-plant) where specialists interact with subsets of generalists' partners [23, 26]. Third, they identify geographic clustering as a factor in the emergence of nestedness, highlighting the role of Silicon Valley as an organisational hub.

Recognising the importance of network position can help investors and entrepreneurs strategise about co-investment partnerships. Policymakers can foster more inclusive and efficient ecosystems by supporting networking platforms, encouraging regional innovation clusters and monitoring concentration risks associated with hub dominance. Furthermore, the fact that nested networks are simultaneously resilient to random failures and vulnerable to the exit of hub investors showcase the need for diversification and succession planning.

It is important tough to consider that this study also has limitations. Crunchbase data, while comprehensive, may under-represent certain sectors or regions [29], and our focus on U.S. companies limits generalisability to other markets. Moreover, the cross-sectional analysis of nestedness, although supplemented by a temporal examination, cannot establish causality between network structure and economic outcomes.

Future research could employ longitudinal models or quasi-experimental designs to examine causal mechanisms, compare investor networks across countries and test whether connectance thresholds for nestedness emergence are universal. Extending the analysis to investor-level nestedness contributions and incorporating dynamic processes such as network rewiring would further deepen understanding of how innovation ecosystems evolve.

In conclusion, this research advances our knowledge on how network structure influences by elucidating the mechanisms through which syndication networks evolve and organise. Untimately, it lays groundwork for creating financial environments that better support diverse innovators and promote sustainable economic growth.

References

[1] Josh Lerner. The syndication of venture capital investments. *Financial Management*, 23(3):16–27, 1994.

[2] Catherine Casamatta and Carole Haritchabalet. Experience, screening and syndication in venture capital investments. *Journal of Financial Intermediation*, 16(3):368–398, 2007.

[3] Sarika Pruthi and Andy Lockett. International venture capital research: From cross-country comparisons to crossing borders. *International Journal of Management Reviews*, 7:135–165, 2005.

[4] Mohamed Benzouai and Khalid Djeffal. A bibliometric review of research on venture capital. *Asian Academy of Management Journal*, 26:47–88, 2021.

[5] Sebastian Mariani. *Nestedness in complex networks*. Springer, 2019.

[6] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.

[7] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009.

[8] Michael Spence. Signaling, screening, and information. In Sherwin Rosen, editor, *Studies in Labor Markets*, pages 319–358. University of Chicago Press, 1981.

[9] Mark Granovetter. Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91(3):481–510, 1985.

[10] Brian Uzzi. Social structure and competition in interfirm networks: The paradox of embeddedness. *Administrative Science Quarterly*, 42(1):35–67, 1997.

[11] Ronald S. Burt. *Structural Holes: The Social Structure of Competition*. Harvard University Press, Cambridge, MA, 1992.

[12] José Vargas-Hernández and Cecilia Morales-Del-Río. Social networks in organizations. *Organicom*, 16, 2019.

[13] Stephen P. Borgatti, Ajay Mehra, Daniel J. Brass, and Giuseppe Labianca. Network analysis in the social sciences. *Science*, 323(5916):892–895, 2009.

[14] Mark E. J. Newman. The structure and function of complex networks. *SIAM Review*, 45(2):167–256, 2003.

[15] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of 'small-world' networks. *Nature*, 393(6684):440–442, 1998.

[16] Carsten Dormann, Jochen Fründ, Nico Blüthgen, and Bernd Gruber. Indices, graphs and null models: Analyzing bipartite ecological networks. *The Open Ecology Journal*, 2:7–24, 2009.

[17] William D. Bygrave. Syndicated investments by venture capital firms: A networking perspective. *Journal of Business Venturing*, 2(2):139–154, 1987.

[18] Stephen P. Borgatti and Daniel S. Halgin. On network theory. *Organization Science*, 22(5):1168–1181, 2011.

[19] Théophile Carniel. *Complex networks in entrepreneurial ecosystems: clustering methodologies and topological structure*. Thèse, Université Paris Cité, 2024.

[20] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008.

[21] Vincent A. Traag, Ludo Waltman, and Nees Jan van Eck. From louvain to leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9(5233):1–12, 2019.

[22] Jean-Michel Dalle, Matthijs Den Besten, and Jeremie Morfin. Accelerator-mediated access to investors among early-stage start-ups. *Annals of Operations Research*, 348(3):1925–1952, 2025.

[23] Jordi Bascompte, Pedro Jordano, and Jens M. Olesen. The nested assembly of plant–animal mutualistic networks. *Proceedings of the National Academy of Sciences*, 100(16):9383–9387, 2003.

[24] Mário Almeida-Neto, Paulo Guimarães, Paulo R. Guimarães Jr, Rafael D. Loyola, and Werner Ulrich. A consistent metric for nestedness analysis in ecological systems: reconciling concept and measurement. *Oikos*, 117(8):1227–1239, 2008.

[25] Werner Ulrich and Nicholas J. Gotelli. Null model analysis of species nestedness patterns. *Ecology*, 88(7):1824–1831, 2007.

[26] Elisa Thébault and Colin Fontaine. Stability of ecological communities and the architecture of mutualistic and trophic networks. *Science*, 329(5993):853–856, 2010.

[27] James A. Brander, Raphael Amit, and Werner Antweiler. Venture-capital syndication: Improved venture selection vs. the value-added hypothesis. *Journal of Economics & Management Strategy*, 11(3):423–452, 2002.

[28] Morten Sorensen. How smart is smart money? a two-sided matching model of venture capital. *The Journal of Finance*, 62(6):2725–2762, 2007.

[29] Jean-Michel Dalle and Matthijs Den Besten. Using crunchbase for economic and managerial research. OECD Science, Technology and Industry Working Papers 2017/08, OECD, November 2017.

[30] Michael J. Barber. Modularity and community detection in bipartite networks. *Physical Review E*, 76(6):066102, 2007.

[31] Amit Bubna, Sanjiv R. Das, and Nagpurnanand Prabhala. Venture capital communities. *The Journal of Financial and Quantitative Analysis*, 55(2):621–651, 2020.

[32] Giovanni Strona, Domenico Nappo, Francesco Boccacci, Simone Fattorini, and Jesús San-Miguel-Ayanz. A fast and unbiased procedure to randomize ecological binary matrices with fixed row and column totals. *Nature Communications*, 5(1):4114, 2014.

[33] Yael V. Hochberg, Alexander Ljungqvist, and Yang Lu. Whom you know matters: Venture capital networks and investment performance. *The Journal of Finance*, 62(1):251–301, 2007.